

Note on Monitored Chern Insulator (Reowrked)

Haoyu Guo

Department of Physics, Cornell University, Ithaca, New York 14853, USA.

I. Bulk model

We consider the following bulk model proposed by Matt:

$$h(k_x, k_y)/t_0 = \sigma^1 \sin k_x + \sigma^2 \sin k_y + \sigma^3 (r - \cos k_x - \cos k_y) . \quad (1)$$

This hamiltonian has a Chern number when $0 < |r| < 2$.

We can write down the real-space tight binding Hamiltonian as

$$h/t_0 = \sigma^1 \frac{T_x^\dagger - T_x}{2i} + \sigma^2 \frac{T_y^\dagger - T_y}{2i} + \sigma^3 \left(r - \frac{T_x + T_x^\dagger + T_y + T_y^\dagger}{2} \right) , \quad (2)$$

where T_x, T_y are the translation operators in the x, y directions respectively.

II. The model on the semi-infinite geometry

Let's write down the second quantized hamiltonian

$$\begin{aligned} H/t_0 = & \sum_{k_x, y=0} c_{k_x, y}^\dagger (\sigma^1 \sin k_x + \sigma^3 (r - \cos k_x)) c_{k_x, y} \\ & + c_{k_x, y}^\dagger \left(\frac{\sigma^2}{2i} - \frac{\sigma^3}{2} \right) c_{k_x, y+1} + c_{k_x, y+1}^\dagger \left(-\frac{\sigma^2}{2i} - \frac{\sigma^3}{2} \right) c_{k_x, y} , \end{aligned} \quad (3)$$

The Green's function is found to be

$$G^{-1}(z, k_x) = z - \sigma^1 \sin k_x - \sigma^3 (r - \cos k_x) - \Sigma(z, k_x) \frac{1 - \sigma^1}{2} , \quad (4)$$

where

$$\Sigma(z, k_x) = \frac{z^2 + 2r \cos k_x - r^2 - \sqrt{z^2 - E_+^2} \sqrt{z^2 - E_-^2}}{2(z - \sin k_x)} . \quad (5)$$

and

$$E_\pm^2 = \sin^2 k_x + (r - \cos k_x \pm 1)^2 . \quad (6)$$

Here the branch cut is the standard one for square root.

The spectral function can be written as the sum of two terms $A = -2\text{Im}G_R = A_{\text{edge}} + A_{\text{bulk}}$.

$$A_{\text{edge}} = 2\pi\delta(z - \sin k_x) \frac{1 - (r - \cos k_x)^2}{2} \theta(1 - (r - \cos k_x)^2) \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} , \quad (7)$$

$$A_{\text{bulk}} = \frac{1}{2(r - \cos k_x)^2} \frac{1}{|z - \sin k_x|} \sqrt{(E_+^2 - z^2)(z^2 - E_-^2)} \theta(E_- < |z| < E_+) \times \begin{pmatrix} (r + z - \cos k_x - \sin k_x)^2 & (r - \cos k_x)^2 - (z - \sin k_x)^2 \\ (r - \cos k_x)^2 - (z - \sin k_x)^2 & (r - z - \cos k_x + \sin k_x)^2 \end{pmatrix}. \quad (8)$$

III. Setting up measurement

We briefly discuss how the measurement is setup. The measurement is described by a set of Kraus operators:

$$\rho \rightarrow K_v \rho K_v^\dagger, \quad (9)$$

with the POVM condition

$$\sum_v K_v^\dagger K_v = \mathbb{1}. \quad (10)$$

We consider measuring a fermion operator

$$\hat{n}_i[\alpha] = (\alpha^\dagger c_i)^\dagger (\alpha^\dagger c_i), \quad (11)$$

where α is a unit vector in the orbital space, so that $\hat{n}_i[\alpha]^2 = \hat{n}_i[\alpha]$, i.e. it has eigenvalue 0,1. Using this property, we can construct the Kraus operator using a v_i^α drawn from the real line.

$$K_{v_i^\alpha} \propto \exp(\kappa v_i^\alpha (\hat{n}_i[\alpha] - 1/2)), \quad \int dv_i^\alpha K_{v_i^\alpha}^\dagger K_{v_i^\alpha} \propto \mathbb{1}. \quad (12)$$

Assuming continuum limit makes sense, the Keldysh action for the measurement takes the form

$$iS_m = \int dt d^d \mathbf{r} \sum_\alpha v^\alpha(t, \mathbf{r}) \sum_{a=1}^r \psi_a^\dagger \alpha \tau^x \alpha^\dagger \psi_a \quad (13)$$

Averaging over the trajectories with $\langle v^\alpha v^\beta \rangle = \gamma^2 \delta^{\alpha\beta}$, we therefore obtain

$$iS_m \Rightarrow \frac{\gamma^2}{2} \int dt d\mathbf{r} \sum_\alpha \psi^\dagger \mathbb{P}_\alpha \tau^x \psi \mathbb{P}_\alpha \tau^x \psi. \quad (14)$$

Here $\mathbb{P}_\alpha = \alpha \alpha^\dagger$ is a projector in the orbital space. Decouple with Green's function $G \propto -i\psi\psi^\dagger$, we obtain

$$iS_m \Rightarrow \frac{\gamma^2}{2} \int_{t,\mathbf{r}} \sum_\alpha \text{Tr}(G \tau^x \mathbb{P}_\alpha G \tau^x \mathbb{P}_\alpha) - f_R \text{Tr}^2(G \tau^x \mathbb{P}_\alpha) \quad (15)$$

The trace is over replica, Keldysh, and orbital indices. To make sure the different Kraus operators commute, we require $\{\mathbb{P}_\alpha\}$ to be orthogonal to each other. For now we ignore the f_R parameter, and just focus on the entanglement sector.

The full action is

$$iS = \ln \det (G_0^{-1} - \Sigma) + \text{Tr} (\Sigma \cdot G) + iS_m \quad (16)$$

We should note that while the saddle point is not very sensitive to the choice of $\{\mathbb{P}_\alpha\}$, but the fluctuations are. In particular, the fluctuations are diagonal in the basis described by $\{\mathbb{P}_\alpha\}$.

IV. Saddle point

The saddle point equation with $f_R = 0$ is

$$\Sigma_M(t, \mathbf{r}) = -\gamma^2 \sum_\alpha \int \frac{d\omega}{2\pi} \frac{dk}{2\pi} \mathbb{P}_\alpha \tau^x G(\omega, k) \tau^x \mathbb{P}_\alpha = \frac{-i\Gamma}{2} \tau_z \sum_\alpha \mathbb{P}_\alpha, \quad (17)$$

Here

$$\Gamma = \gamma^2, \quad (18)$$

and we have evaluated the ω -integral using the sum rule (leading to a result proportional identity), and assume the BZ spans $k \in [-\pi, \pi]$.

V. Orbit monitoring

Now we consider the orbit monitoring case, where $\{\mathbb{P}_\alpha\} = \{\mathbb{P}_{z,-}, \mathbb{P}_{z,+}\}$, i.e. the eigenbasis of σ_z . In this setting, the saddle point self-energy is

$$\Sigma_{M,\text{saddle}} = -\frac{i\Gamma}{2} \tau_z. \quad (19)$$

We next derive the NLSM for the fluctuations around the saddle point. We first integrate out G , so the action becomes

$$iS = \ln \det (G_0^{-1} - \Sigma) + \frac{1}{2} \text{Tr} (\Sigma \cdot W_G^{-1}[\Sigma]) . \quad (20)$$

Here, the functional W_G is local in space time, and the local action is given by

$$W_G[F] = -\gamma^2 \sum_\alpha [\tau^x \mathbb{P}_\alpha F \mathbb{P}_\alpha \tau^x - \tau^x \mathbb{P}_\alpha f_R \text{Tr} (F \tau^x \mathbb{P}_\alpha)] \quad (21)$$

A. Choice of the parameter f_R

In this subsection, we determine the value of f_R using the requirement that Σ_M can be extended to a causal solution

$$\Sigma_M = \begin{pmatrix} \Sigma_R & \Sigma_K \\ 0 & \Sigma_A \end{pmatrix}, \quad (22)$$

we assume all of them remain to be a frequency-independent number. Substituting the above ansatz into the saddle point equation, we obtain

$$\sum_{\alpha} \mathbb{P}_{\alpha} G_K \mathbb{P}_{\alpha} = \sum_{\alpha} f_R \mathbb{P}_{\alpha} \text{Tr} (\mathbb{P}_{\alpha} G_K). \quad (23)$$

Here G_K is the local Keldysh Green's function of the saddle. The above equation simply says that the orbital-diagonal entries of G_K is related to its trace over the replica indices, and therefore it requires

$$f_R = \frac{1}{R}, \quad (24)$$

which is the same conclusion as the previous AIII paper. The remaining determines the Keldysh self-energy Σ_K in terms of the local Keldysh G_K . In the upper-triangular Keldysh convention used throughout this note, Σ_K is anti-Hermitian, so the Hermitian density kernel carries an additional factor of $-i$:

$$\Sigma_K = -i\gamma^2 \begin{pmatrix} G_K^1 & 0 \\ 0 & G_K^2 \end{pmatrix}_{\text{orbit}} = -i\Gamma \begin{pmatrix} 1 - 2n_1 & 0 \\ 0 & 1 - 2n_2 \end{pmatrix}, \quad (25)$$

i.e. the Keldysh component is an anti-Hermitian matrix set by the orbital densities, while the Hermitian kernel is $\Gamma(1 - 2n_a)$.

B. Deriving the NLSM

We again consider the half-filling case, so that the saddle has $\Sigma_K = 0$. Before choosing coordinates on the soft manifold, it is useful to spell out the orbital structure of Q . The monitoring projectors are $\mathbb{P}_{z,+}$ and $\mathbb{P}_{z,-}$, so the most general local measurement saddle within the orbital-density Hubbard-Stratonovich channel is block diagonal in this monitored orbital basis:

$$Q_{\text{orb}} = \mathbb{P}_{z,+} Q_+ + \mathbb{P}_{z,-} Q_-, \quad Q_{\pm}^2 = \mathbb{1}. \quad (26)$$

Here $Q_{\pm}$ are matrices in Keldysh and replica space. Equivalently,

$$Q_{\pm} = T_{\pm}^{-1} \tau^z T_{\pm}$$

with the usual causal constraints on $T_{\pm}$. In the explicit coordinates used below one may write

$$Q_{\pm} = \cos \phi_{\pm} [\tau^z \cos B_{\pm} + \tau^y \sin B_{\pm}] + \sin \phi_{\pm} \tau^x, \quad B_{\pm} = \theta_{\pm} \mathbb{1}_R + Y_{\pm}, \quad (27)$$

with $Y_{\pm}^{\dagger} = Y_{\pm}$ and $\text{Tr}_R Y_{\pm} = 0$. Since the two orbital projectors are orthogonal, (26) satisfies

$$Q_{\text{orb}}^2 = \mathbb{P}_{z,+} Q_+^2 + \mathbb{P}_{z,-} Q_-^2 = \mathbb{1}$$

whenever $Q_{\pm}^2 = \mathbb{1}$.

Equation (28) below is the diagonal submanifold

$$Q_+ = Q_- \equiv Q, \quad Q_{\text{orb}} = \mathbb{1}_{\text{orb}} Q.$$

This is the unique gapless orbital sector. The reason is that only a common Keldysh-replica rotation of both monitored orbitals is a symmetry of the full fermion determinant. A relative orbital rotation $Q_+ \neq Q_-$ inserts the orbital matrix $\mathbb{P}_{z,+} - \mathbb{P}_{z,-} = \sigma_z$ at the fluctuation vertex. The single-particle Hamiltonian contains orbital matrices that do not commute with σ_z , for example the $\sin q \sigma_x$ term, so this relative rotation is not protected by a Ward identity. In the quadratic action the common mode has the subtracted bubble $\Pi_{RA}(\Omega, k) - \Pi_{RA}(0, 0)$, which vanishes at zero external arguments, whereas the relative orbital mode involves bubbles with σ_z insertions, such as

$$\Pi_{RA}^{zz}(0, 0) = \int \frac{d\omega dq}{(2\pi)^2} \text{tr}_{\text{orb}} [\sigma_z G_R(\omega, q) \sigma_z G_A(\omega, q)],$$

and this does not cancel the local measurement term. Thus $Q_+ - Q_-$ is a massive orbital-coherence mode. Off-diagonal orbital components, which would mix $\mathbb{P}_{z,+}$ and $\mathbb{P}_{z,-}$, are even more directly outside the monitored-density saddle channel and are also massive. The NLSM therefore keeps only the orbital-identity component.

It is convenient to rewrite this remaining fluctuation manifold in a form that makes the nonlinear constraint manifest. Let

$$\Sigma = -\frac{i\Gamma}{2} Q, \quad Q = \cos \phi [\tau^z \cos B + \tau^y \sin B] + \sin \phi \tau^x, \quad (28)$$

where

$$B = \theta \mathbb{1}_R + Y, \quad Y^\dagger = Y, \quad \text{Tr}_R Y = 0. \quad (29)$$

This is equivalent to the previous parametrization with $\mathcal{X} = e^{iB} = e^{i\theta} X$, $X \in SU(R)$. Since $\cos B$ and $\sin B$ commute, one immediately checks that $Q^2 = \mathbb{1}$. The saddle point corresponds to

$$Q_{\text{saddle}} = \tau^z, \quad \Sigma_{\text{saddle}} = -\frac{i\Gamma}{2} \tau^z. \quad (30)$$

For small fluctuations,

$$Q = \tau^z + B \tau^y + \phi \tau^x - \frac{1}{2} (B^2 + \phi^2) \tau^z + O(\Phi^3), \quad (31)$$

so the linear fluctuation of the self-energy is

$$\delta\Sigma = -\frac{i\Gamma}{2} (B \tau^y + \phi \tau^x) + O(\Phi^2). \quad (32)$$

Expanding the action around the saddle gives

$$iS = iS_{\text{saddle}} + \frac{1}{2} \text{Tr} (\delta\Sigma \cdot W_G^{-1} [\delta\Sigma]) - \frac{1}{2} \text{Tr} (G_{\text{saddle}} \delta\Sigma G_{\text{saddle}} \delta\Sigma) + O(\Phi^3). \quad (33)$$

The linear term vanishes by the saddle-point equation. Moreover, the parametrization (28) moves along the degenerate local saddle manifold, so the purely local part of the quadratic kernel cancels between the two terms in (33). The surviving quadratic action is therefore controlled by the nonlocal fermion bubble.

We now Fourier transform the fluctuation fields and use the shorthand

$$\int_Q \equiv \int \frac{d\Omega dk}{(2\pi)^2}. \quad (34)$$

At linear order, the B mode couples through τ^y and the ϕ mode through τ^x . Using

$$G(\omega, k) = \begin{pmatrix} G_R(\omega, k) & G_K(\omega, k) \\ 0 & G_A(\omega, k) \end{pmatrix}, \quad G_A = G_R^\dagger, \quad (35)$$

the relevant Keldysh traces are

$$\text{tr}_K(G_1 \tau^y G_2 \tau^y) = \text{tr}_{\text{orb}}(G_{R,1} G_{A,2} + G_{A,1} G_{R,2} - G_{K,1} G_{K,2}), \quad (36)$$

$$\text{tr}_K(G_1 \tau^x G_2 \tau^x) = \text{tr}_{\text{orb}}(G_{R,1} G_{A,2} + G_{A,1} G_{R,2} + G_{K,1} G_{K,2}), \quad (37)$$

$$\text{tr}_K(G_1 \tau^x G_2 \tau^y) = i \text{tr}_{\text{orb}}(G_{R,1} G_{A,2} - G_{A,1} G_{R,2} + G_{K,1} G_{K,2}). \quad (38)$$

There is therefore a genuine mixed B - ϕ quadratic kernel. Since the saddle Green's function is proportional to the identity in replica space, the $U(1)$ and $SU(R)$ parts of B share the same diagonal kernel, but only the replica-singlet θ mode can mix with ϕ :

$$\text{Tr}_R[B(-Q)B(Q)] = R\theta(-Q)\theta(Q) + \text{Tr}_R[Y(-Q)Y(Q)]. \quad (39)$$

while

$$\text{Tr}_R[B(-Q)\phi(Q)] = R\theta(-Q)\phi(Q), \quad \text{Tr}_R[Y(-Q)\phi(Q)] = 0. \quad (40)$$

Define the retarded-advanced bubble

$$\Pi_{RA}(\Omega, k) = \int \frac{d\omega dq}{(2\pi)^2} \text{tr}_{\text{orb}}[G_R(\omega + \Omega, q + k) G_A(\omega, q)], \quad (41)$$

and the Keldysh-Keldysh bubble

$$\Pi_{KK}(\Omega, k) = \int \frac{d\omega dq}{(2\pi)^2} \text{tr}_{\text{orb}}[G_K(\omega + \Omega, q + k) G_K(\omega, q)]. \quad (42)$$

At the quadratic level, the two kernels entering the NLSM are

$$\begin{aligned} \mathcal{W}_B(\Omega, k) &= [\Pi_{RA}(\Omega, k) + \Pi_{RA}(-\Omega, -k) - 2\Pi_{RA}(0, 0)] \\ &\quad - [\Pi_{KK}(\Omega, k) - \Pi_{KK}(0, 0)], \end{aligned} \quad (43)$$

$$\begin{aligned} \mathcal{W}_\phi(\Omega, k) &= [\Pi_{RA}(\Omega, k) + \Pi_{RA}(-\Omega, -k) - 2\Pi_{RA}(0, 0)] \\ &\quad + [\Pi_{KK}(\Omega, k) - \Pi_{KK}(0, 0)]. \end{aligned} \quad (44)$$

The off-diagonal kernels are

$$\mathcal{W}_{\theta\phi}(\Omega, k) = i[\Pi_{RA}(\Omega, k) - \Pi_{RA}(-\Omega, -k) + \Pi_{KK}(\Omega, k) - \Pi_{KK}(0, 0)], \quad (45)$$

$$\mathcal{W}_{\phi\theta}(\Omega, k) = i[\Pi_{RA}(-\Omega, -k) - \Pi_{RA}(\Omega, k) + \Pi_{KK}(\Omega, k) - \Pi_{KK}(0, 0)]. \quad (46)$$

Therefore

$$iS^{(2)} = -\frac{\Gamma^2}{8} \int_Q \left[\mathcal{W}_B(\Omega, k) \left(R\theta(-Q)\theta(Q) + \text{Tr}_R[Y(-Q)Y(Q)] \right) + R\mathcal{W}_{\theta\phi}(\Omega, k)\theta(-Q)\phi(Q) + R\mathcal{W}_{\phi\theta}(\Omega, k)\phi(-Q)\theta(Q) \right] \quad (47)$$

The retarded-advanced part is the same fermion bubble analyzed in Appendix A 1. In the notation introduced there,

$$\Gamma \text{Re} \left[\frac{\Pi_{RA}(\Omega, k) + \Pi_{RA}(-\Omega, -k)}{2} - \Pi_{RA}(0, 0) \right] = \Delta(\Omega, k), \quad (48)$$

so the real part of the retarded-advanced contribution to the NLSM kernel is

$$\text{Re } \mathcal{W}_{B,\phi}^{RA}(\Omega, k) = \frac{2}{\Gamma} \Delta(\Omega, k). \quad (49)$$

For the off-diagonal piece, the retarded-advanced contribution is the odd part

$$\mathcal{W}_{\theta\phi}^{RA}(\Omega, k) = i \left[\Pi_{RA}(\Omega, k) - \Pi_{RA}(-\Omega, -k) \right], \quad (50)$$

so the θ - ϕ coupling starts at odd order in the external variables. Moreover,

$$\Pi_{KK}(-\Omega, -k) = \Pi_{KK}(\Omega, k), \quad (51)$$

which follows directly from the definition (42) by shifting the integration variables and using cyclicity of the orbital trace. Therefore the Keldysh-Keldysh contribution to the mixed kernel is even in (Ω, k) and, after subtracting the local piece $\Pi_{KK}(0, 0)$, it starts only at even order. In particular, the leading odd small- (Ω, k) term remains proportional to Ωk and comes entirely from the retarded-advanced sector. Hence, if we leave the Keldysh-Keldysh piece abstract for the moment, the quadratic NLSM inherits the previously established small- (Ω, k) structure

$$\Delta(\Omega, k) = -A_k k^2 + A_{\Omega k} \Omega k - \frac{\Omega^2}{8\pi} \log \frac{C_\Omega}{\Gamma|\Omega|} + \dots \quad (52)$$

The $U(1)$ mode θ and the $SU(R)$ mode Y therefore share the same singular retarded-advanced diagonal kernel, the ϕ mode differs by the sign of the abstract Π_{KK} contribution, and there is in general a θ - ϕ mixing kernel $\mathcal{W}_{\theta\phi}$ as well.

1. Diagonalizing the θ - ϕ sector for $\Pi_{KK} = 0$

If we temporarily set

$$\Pi_{KK}(\Omega, k) = 0, \quad (53)$$

then the singlet θ - ϕ block of (47) takes the form

$$\begin{aligned} iS_{\theta\phi}^{(2)} = & -\frac{\Gamma^2 R}{8} \int_Q \left[W_s(\Omega, k) \theta(-Q) \theta(Q) + W_s(\Omega, k) \phi(-Q) \phi(Q) \right. \\ & \left. + W_a(\Omega, k) \theta(-Q) \phi(Q) - W_a(\Omega, k) \phi(-Q) \theta(Q) \right], \end{aligned} \quad (54)$$

where

$$W_s(\Omega, k) = \Pi_{RA}(\Omega, k) + \Pi_{RA}(-\Omega, -k) - 2\Pi_{RA}(0, 0), \quad (55)$$

$$W_a(\Omega, k) = i \left[\Pi_{RA}(\Omega, k) - \Pi_{RA}(-\Omega, -k) \right]. \quad (56)$$

Now define the circular combinations

$$\psi_{\pm}(Q) = \frac{\theta(Q) \pm i\phi(Q)}{\sqrt{2}}, \quad \bar{\psi}_{\pm}(Q) = \frac{\theta(-Q) \mp i\phi(-Q)}{\sqrt{2}}. \quad (57)$$

In terms of these variables,

$$\begin{aligned} \bar{\psi}_+(Q)\psi_+(Q) &= \frac{1}{2} \left[\theta(-Q)\theta(Q) + \phi(-Q)\phi(Q) \right. \\ &\quad \left. + i\theta(-Q)\phi(Q) - i\phi(-Q)\theta(Q) \right], \end{aligned} \quad (58)$$

$$\begin{aligned} \bar{\psi}_-(Q)\psi_-(Q) &= \frac{1}{2} \left[\theta(-Q)\theta(Q) + \phi(-Q)\phi(Q) \right. \\ &\quad \left. - i\theta(-Q)\phi(Q) + i\phi(-Q)\theta(Q) \right]. \end{aligned} \quad (59)$$

Therefore the singlet block diagonalizes as

$$iS_{\theta\phi}^{(2)} = -\frac{\Gamma^2 R}{8} \int_Q \left[(W_s - iW_a) \bar{\psi}_+(Q)\psi_+(Q) + (W_s + iW_a) \bar{\psi}_-(Q)\psi_-(Q) \right]. \quad (60)$$

Using (55) and (56), the two eigen-kernels are simply

$$W_s(\Omega, k) - iW_a(\Omega, k) = 2 \left[\Pi_{RA}(\Omega, k) - \Pi_{RA}(0, 0) \right], \quad (61)$$

$$W_s(\Omega, k) + iW_a(\Omega, k) = 2 \left[\Pi_{RA}(-\Omega, -k) - \Pi_{RA}(0, 0) \right]. \quad (62)$$

So when $\Pi_{KK} = 0$, the θ - ϕ sector is exactly diagonalized by the circular combinations $\psi_{\pm}$: one mode is governed by the retarded- advanced bubble at (Ω, k) , and the other by the same bubble at $(-\Omega, -k)$. The mixing seen in the (θ, ϕ) basis is therefore just the statement that the natural eigenmodes are $\theta \pm i\phi$, rather than θ and ϕ separately.

2. Including the even Π_{KK} contribution

When $\Pi_{KK} \neq 0$, the symmetry (51) implies that only the even combination

$$K_s(\Omega, k) \equiv \Pi_{KK}(\Omega, k) - \Pi_{KK}(0, 0) \quad (63)$$

can enter the singlet block. Then (54) generalizes to

$$iS_{\theta\phi}^{(2)} = -\frac{\Gamma^2 R}{8} \int_Q \left[W_s \theta(-Q)\theta(Q) + W_s \phi(-Q)\phi(Q) + W_a \theta(-Q)\phi(Q) - W_a \phi(-Q)\theta(Q) \right. \\ \left. - K_s \theta(-Q)\theta(Q) + K_s \phi(-Q)\phi(Q) + iK_s \theta(-Q)\phi(Q) + iK_s \phi(-Q)\theta(Q) \right]. \quad (64)$$

Transforming to the circular variables $\psi_{\pm}$ gives

$$iS_{\theta\phi}^{(2)} = -\frac{\Gamma^2 R}{8} \int_Q \left[(W_s - iW_a) \bar{\psi}_+(Q)\psi_+(Q) + (W_s + iW_a) \bar{\psi}_-(Q)\psi_-(Q) - 2K_s(\Omega, k) \bar{\psi}_+(Q)\psi_-(Q) \right]. \quad (65)$$

Using (55) and (56), this can be written as

$$iS_{\theta\phi}^{(2)} = -\frac{\Gamma^2 R}{4} \int_Q \left[(\Pi_{RA}(\Omega, k) - \Pi_{RA}(0, 0)) \bar{\psi}_+(Q)\psi_+(Q) \right. \\ \left. + (\Pi_{RA}(-\Omega, -k) - \Pi_{RA}(0, 0)) \bar{\psi}_-(Q)\psi_-(Q) - K_s(\Omega, k) \bar{\psi}_+(Q)\psi_-(Q) \right]. \quad (66)$$

So the effect of a nonzero Π_{KK} is not to change the odd Ωk coefficient of the diagonalized retarded-advanced kernels. Instead, it introduces an even off-diagonal mixing between the two circular modes ψ_+ and ψ_- . The simple diagonalization by $\theta \pm i\phi$ therefore holds only when $\Pi_{KK} = 0$; otherwise those combinations remain useful because they isolate the retarded-advanced odd part, but they are coupled to each other by the even kernel $K_s(\Omega, k)$.

3. Low-energy dispersion of the (θ, ϕ) sector

To identify the low-energy singlet modes, we now follow the same strategy as in the Guo et al. reference in the **References/** folder: first diagonalize the (θ, ϕ) block in the circular basis, then insert the low-energy bubble expansion, and finally read off the resulting soft trajectories and pole structure.

As discussed above, the full KK bubble is smooth at the origin and extremely small numerically; Appendix A 3 gives

$$\text{Re}\Pi_{KK}(0, 0) \approx -8.38 \times 10^{-5}, \quad -8.08 \times 10^{-6}, \quad -6.70 \times 10^{-7} \quad (\Gamma = 20, 40, 80),$$

consistent with the large- Γ estimate $\Pi_{KK} = O(\Gamma^{-4})$; see (A59). If we therefore treat Π_{KK} as independent of (Ω, k) at low energy, then $K_s(\Omega, k) = \Pi_{KK}(\Omega, k) - \Pi_{KK}(0, 0) \approx 0$, and the singlet block is controlled entirely by the retarded-advanced kernel.

Using (60), the quadratic action becomes

$$iS_{\theta\phi}^{(2)} \simeq -\frac{\Gamma^2 R}{4} \int_Q \left[\delta\Pi_{RA}(\Omega, k) \bar{\psi}_+(Q) \psi_+(Q) + \delta\Pi_{RA}(-\Omega, -k) \bar{\psi}_-(Q) \psi_-(Q) \right], \quad (67)$$

where

$$\delta\Pi_{RA}(\Omega, k) \equiv \Pi_{RA}(\Omega, k) - \Pi_{RA}(0, 0).$$

The appendix result (A34) then implies, to leading order in small external arguments,

$$\Gamma \operatorname{Re} \delta\Pi_{RA}(\pm\Omega, \pm k) = -A_k k^2 \pm A_{\Omega k} \Omega k - \frac{\Omega^2}{8\pi} \log \frac{C_\Omega}{\Gamma|\Omega|} + \dots, \quad (68)$$

with

$$A_k \xrightarrow{\Gamma \rightarrow \infty} \frac{1}{3\sqrt{3}} - \frac{1}{8}, \quad A_{\Omega k} \xrightarrow{\Gamma \rightarrow \infty} \frac{1}{3\sqrt{3}}. \quad (69)$$

The even real part is not the whole story, however. The full complex bubble can also contain terms linear in Ω and k . As shown in Appendix A1b, those linear terms are necessarily purely imaginary:

$$\Gamma \delta\Pi_{RA}(\Omega, k) = i[b_\Omega(\Gamma) \Omega - b_k(\Gamma) k] - A_k k^2 + A_{\Omega k} \Omega k - \frac{\Omega^2}{8\pi} \log \frac{C_\Omega}{\Gamma|\Omega|} + \dots, \quad (70)$$

with

$$b_\Omega(\Gamma) = O(\Gamma^{-1}), \quad b_k(\Gamma) = O(\Gamma^{-1}). \quad (71)$$

These odd analytic terms therefore do not modify the real-part zero condition used below to identify the soft real-frequency trajectories, although they do contribute to the phase of the circular $\psi_\pm$ kernels and hence to the θ - ϕ mixing structure. Since Ω in (41) is already the physical real frequency, there is no additional analytic continuation here. The low-energy dispersion is read off directly from the real-frequency quadratic kernels of $\psi_\pm$.

It is convenient to define the dimensionless ratio

$$v \equiv \frac{A_{\Omega k}}{A_k}, \quad v \xrightarrow{\Gamma \rightarrow \infty} \frac{1/(3\sqrt{3})}{1/(3\sqrt{3}) - 1/8} \approx 2.853. \quad (72)$$

Then (68) becomes

$$\Gamma \operatorname{Re} \delta\Pi_{RA}(\pm\Omega, \pm k) = -A_k \left[k^2 \mp v \Omega k + \frac{\Omega^2}{8\pi A_k} \log \frac{C_\Omega}{\Gamma|\Omega|} \right] + \dots. \quad (73)$$

If one temporarily drops the nonanalytic $\Omega^2 \log |\Omega|$ term, the two circular modes would have the linear dispersions

$$\Omega_\pm(k) \simeq \pm \frac{k}{v}, \quad \frac{1}{v} \approx 0.350. \quad (74)$$

This is the direct analogue of the shear structure found in the Guo et al. analysis: the ψ_+ and ψ_- sectors disperse in opposite directions.

Keeping the logarithmic term, the zero of the real part of the quadratic kernel obeys

$$k^2 \mp v \Omega k + \frac{\Omega^2}{8\pi A_k} \log \frac{C_\Omega}{\Gamma|\Omega|} = 0. \quad (75)$$

For fixed Ω , this is a quadratic equation for k :

$$k_{\pm,\sigma}(\Omega) = \frac{\Omega}{2A_k} \left[\pm A_{\Omega k} + \sigma \sqrt{A_{\Omega k}^2 - \frac{A_k}{2\pi} \log \frac{C_\Omega}{\Gamma|\Omega|}} \right], \quad \sigma = \pm. \quad (76)$$

Real solutions exist only when

$$A_{\Omega k}^2 > \frac{A_k}{2\pi} \log \frac{C_\Omega}{\Gamma|\Omega|}. \quad (77)$$

Because the logarithm grows without bound as $|\Omega| \rightarrow 0$, this condition eventually fails at asymptotically low frequency. Therefore the (θ, ϕ) sector does not possess an exact real-frequency pole that survives all the way to the origin. Instead, the linear relation $\Omega_\pm(k) \simeq \pm k/v$ describes the incipient dispersion that is progressively cut off by the singular $\Omega^2 \log(1/|\Omega|)$ term.

Even when the discriminant is negative, the kernel still has a soft ridge. Completing the square in (73) gives

$$k^2 \mp v \Omega k + \frac{\Omega^2}{8\pi A_k} \log \frac{C_\Omega}{\Gamma|\Omega|} = \left(k \mp \frac{v}{2} \Omega \right)^2 + \left[\frac{1}{8\pi A_k} \log \frac{C_\Omega}{\Gamma|\Omega|} - \frac{v^2}{4} \right] \Omega^2. \quad (78)$$

Thus the minimum of the real-frequency kernel at fixed Ω lies along

$$k_\pm^*(\Omega) = \pm \frac{v}{2} \Omega, \quad (79)$$

while the residual stiffness along this ridge is set by the bracketed Ω^2 coefficient in (78). At sufficiently small $|\Omega|$, the logarithm dominates, so the ridge remains soft but does not turn into a true zero mode away from $(\Omega, k) = (0, 0)$.

C. Implications of the bubble analysis

The technical analysis of the fermion bubble is collected in Appendix A. Here we only summarize the conclusions that enter the NLSM.

First, the retarded-advanced bubble controls the singular low-energy structure. In the notation of (48),

$$\Delta(\Omega, k) = -A_k k^2 + A_{\Omega k} \Omega k - \frac{\Omega^2}{8\pi} \log \frac{C_\Omega}{\Gamma|\Omega|} + \dots, \quad (80)$$

with

$$A_k \xrightarrow{\Gamma \rightarrow \infty} \frac{1}{3\sqrt{3}} - \frac{1}{8}, \quad A_{\Omega k} \xrightarrow{\Gamma \rightarrow \infty} \frac{1}{3\sqrt{3}}. \quad (81)$$

These coefficients come from the exact $--$ sector of the full Green's function rather than from the bare edge pole alone; see Appendix A 1.

Second, the Keldysh-Keldysh bubble is even under $(\Omega, k) \rightarrow (-\Omega, -k)$ and therefore does not affect the leading odd Ωk term. More importantly, the full KK bubble is smooth near the origin and is strongly suppressed at large Γ ; see Appendix A 3. Thus the nonanalytic low-energy structure of the quadratic NLSM kernel comes entirely from the retarded-advanced sector.

D. Replica sector Y

The traceless replica field Y is simpler than the singlet sector because it does not mix with ϕ . From (47), its quadratic action is

$$iS_Y^{(2)} = -\frac{\Gamma^2}{8} \int_Q \mathcal{W}_Y(\Omega, k) \text{Tr}_R[Y(-Q)Y(Q)], \quad (82)$$

with

$$\mathcal{W}_Y(\Omega, k) \equiv \mathcal{W}_B(\Omega, k) = [\Pi_{RA}(\Omega, k) + \Pi_{RA}(-\Omega, -k) - 2\Pi_{RA}(0, 0)] - [\Pi_{KK}(\Omega, k) - \Pi_{KK}(0, 0)]. \quad (83)$$

So the Y kernel is exactly the same diagonal τ^y kernel as the replica-singlet field θ , but without any coupling to ϕ .

Because the retarded-advanced bubble carries the singular small- (Ω, k) dependence while the full KK bubble is smooth and strongly suppressed, the leading low-energy behavior of the Y sector is the same as that of the θ sector. In particular, the dominant momentum, mixed, and frequency dependence are inherited from (80).

VI. Generic- r Fermion-Bubble Coefficients

The preceding fermion-bubble analysis was written at the symmetric value $r = 1$. Here we keep the same Chern-insulator phase,

$$0 < r < 2, \quad (84)$$

and redo the large- Γ coefficient extraction. To avoid overloading the same symbol in several roles, this section uses the following conventions: r is the microscopic mass parameter of the Chern-insulator Hamiltonian, q_r is the positive momentum at which the edge mode touches the bulk continuum, c_r is the edge velocity at that touching point, and s_r is the corresponding sine. Explicitly,

$$q_r \equiv \arccos(r - 1), \quad c_r \equiv r - 1, \quad s_r \equiv \sin q_r = \sqrt{r(2 - r)}. \quad (85)$$

Thus c_r is not a new coupling: it is simply $\cos q_r$, or equivalently the endpoint value of the group velocity $\partial_q \sin q$. The main change relative to $r = 1$ is that the edge-bulk touching points move from $\pm\pi/2$ to $\pm q_r$. The edge pole exists on the interval

$$\mathcal{I}_r = (-q_r, q_r), \quad |r - \cos q| < 1, \quad (86)$$

and its residue is

$$Z_r(q) \equiv 1 - (r - \cos q)^2. \quad (87)$$

Inside $\mathcal{I}_r$, the bulk self-energy has the pole

$$\Sigma_R(\omega, q) = \frac{Z_r(q)}{\omega - \sin q + i0} + \Sigma_{\text{reg}}(\omega, q), \quad (88)$$

while outside $\mathcal{I}_r$ this pole is absent. In the σ_x eigenbasis, the retarded inverse Green's function is

$$G_R^{-1}(\omega, q) = \begin{pmatrix} \omega - \sin q + i\Gamma/2 & -(r - \cos q) \\ -(r - \cos q) & \omega + \sin q - \Sigma_R(\omega, q) + i\Gamma/2 \end{pmatrix}. \quad (89)$$

As before, the large- Γ small-argument coefficients are dominated by the $--$ component. On the resonance strip

$$\omega - \sin q = O\left(\frac{Z_r(q)}{\Gamma}\right), \quad (90)$$

the leading local form is

$$G_{--}^{\text{loc}}(\omega, q) = \frac{\omega - \sin q}{i\Gamma(\omega - \sin q)/2 - Z_r(q)}. \quad (91)$$

For the reduced local bubble, the ω integral is again elementary. After taking the real part and subtracting the local $(\Omega, k) = (0, 0)$ piece, the small- (Ω, k) density is

$$\Gamma \text{Re } \mathcal{I}_\Gamma(q; \Omega, k) = -\frac{4Z_r(q)}{\Gamma^2} - \frac{(\Omega - k \cos q)^2}{4Z_r(q)} + O\left(Q^3, \frac{Q^2}{\Gamma^2}, \frac{k}{\Gamma^2}\right), \quad (92)$$

where Q denotes either external argument. Therefore the formal local Taylor kernel is

$$\Delta_r^{(--)}(\Omega, k) = -\int_{-q_r}^{q_r} \frac{dq}{8\pi} \frac{(\Omega - k \cos q)^2}{Z_r(q)} + \dots. \quad (93)$$

At $r = 1$, the factors of $\cos q$ in the k^2 and Ωk terms remove the endpoint divergence, giving the finite constants quoted above. For a generic $r \neq 1$, however, the endpoint velocity is $\cos q_r = c_r \neq 0$, so the same endpoint that produced the $\Omega^2 \log |\Omega|$ term now also produces logarithms in the momentum and mixed coefficients.

Near either touching point,

$$Z_r(q_r - p) = 2s_r p + O(p^2), \quad Z_r(-q_r + p) = 2s_r p + O(p^2), \quad (94)$$

and $\cos q \rightarrow c_r$. Thus the singular endpoint contribution is

$$\Delta_{r,\text{sing}}^{(--)}(\Omega, k) = -\frac{(\Omega - c_r k)^2}{8\pi s_r} \log \frac{C_r}{\Gamma|\Omega - c_r k|} + O(Q^2), \quad (95)$$

where Q denotes a generic small external argument, either Ω or k . The combination $\Omega - c_r k$ is the mismatch between the external frequency and the edge dispersion linearized at either touching point in the bubble convention used above:

$$\Omega + \sin(q - k) - \sin q \simeq \Omega - c_r k \quad (q \simeq \pm q_r).$$

The constant C_r is a nonuniversal matching constant. It absorbs the finite part of the pure endpoint integral proportional to $(\Omega - c_r k)^2$, and it also depends on the precise convention used to match the local endpoint model onto the nonsingular part of the full lattice Green's function. The important universal statement in (95) is therefore the coefficient of the logarithm.

The infrared cutoff in this logarithm is fixed by the point where the fixed- q Taylor expansion (93) stops being valid. Near the right touching point, $q = q_r - p$, the residue is

$$Z_r(q) \simeq 2s_r p.$$

The unexpanded local kernel contains the denominator

$$[Z_r(q) + Z_r(q - k)]^2 + \frac{\Gamma^2}{4} [\Omega + \sin(q - k) - \sin q]^2.$$

Thus the Taylor expansion in external arguments assumes

$$Z_r(q) \gg \Gamma |\Omega - c_r k|$$

up to order-one constants. It fails when

$$p \lesssim p_{\min}, \quad p_{\min} \sim \frac{\Gamma |\Omega - c_r k|}{2s_r}.$$

The endpoint integral therefore has the structure

$$\int_{p_{\min}}^{p_{\text{UV}}} \frac{dp}{p} = \log \frac{p_{\text{UV}}}{p_{\min}} = \log \frac{C_r}{\Gamma |\Omega - c_r k|},$$

where the upper matching scale and all order-one constants have been absorbed into C_r . When $\Omega \simeq c_r k$, the leading endpoint mismatch is tuned away and subleading terms in the external expansion regulate the kernel; along that direction one should use the unexpanded endpoint kernel rather than only (95).

With this convention for C_r , the remaining regular endpoint term is unambiguous and comes from the convergent difference

$$\Delta_{r,\text{reg}}^{(--)}(\Omega, k) = -\frac{1}{8\pi} \int_{-q_r}^{q_r} dq \frac{(\Omega - k \cos q)^2 - (\Omega - c_r k)^2}{Z_r(q)}. \quad (96)$$

Using

$$Z_r(q) = (\cos q - c_r)(r + 1 - \cos q),$$

the numerator of the subtracted expression factorizes as

$$(\Omega - k \cos q)^2 - (\Omega - c_r k)^2 = (\cos q - c_r) [-2\Omega k + k^2(\cos q + c_r)].$$

This cancels the endpoint zero in $Z_r(q)$. Therefore the integrand in (96) is nonsingular:

$$\frac{(\Omega - k \cos q)^2 - (\Omega - c_r k)^2}{Z_r(q)} = \frac{-2\Omega k + k^2(\cos q + c_r)}{r + 1 - \cos q}. \quad (97)$$

Define

$$J_r \equiv \int_{-q_r}^{q_r} \frac{dq}{r + 1 - \cos q} = \frac{4}{\sqrt{r(r+2)}} \arccos \frac{r}{2}. \quad (98)$$

Then the regular endpoint contribution is

$$\Delta_{r,\text{reg}}^{(--)}(\Omega, k) = B_{\Omega k}^{\text{reg}}(r) \Omega k - A_k^{\text{reg}}(r) k^2, \quad (99)$$

with

$$B_{\Omega k}^{\text{reg}}(r) = \frac{J_r}{4\pi}, \quad A_k^{\text{reg}}(r) = \frac{rJ_r - q_r}{4\pi}. \quad (100)$$

There is no separate Ω^2 regular coefficient in this convention, because the finite analytic term proportional to $(\Omega - c_r k)^2$ has been absorbed into C_r . Choosing a different C_r shifts the coefficients of Ω^2 , Ωk , and k^2 in the fixed ratio carried by $(\Omega - c_r k)^2$, but it does not change (96)–(100).

Combining the singular and regular endpoint pieces in this reduced endpoint convention gives the local large- Γ result

$$\Delta_r^{(--)}(\Omega, k) = -\frac{(\Omega - c_r k)^2}{8\pi s_r} \log \frac{C_r^{\text{end}}}{\Gamma|\Omega - c_r k|} + B_{\Omega k}^{\text{reg}}(r) \Omega k - A_k^{\text{reg}}(r) k^2 + \dots \quad (101)$$

The first term has been written with C_r^{end} to emphasize that this constant belongs to the reduced endpoint model with (99) kept separately. It can be computed analytically. The cleanest definition is obtained on the $k = 0$ line, where the unexpanded finite- Γ kernel is

$$\Delta_{\text{end}}(\Omega, 0) = -\frac{\Omega^2}{8\pi} \int_{-q_r}^{q_r} dq \frac{Z_r(q)}{Z_r(q)^2 + a^2}, \quad a \equiv \frac{\Gamma|\Omega|}{4}. \quad (102)$$

Define

$$I_r(a) \equiv \int_{-q_r}^{q_r} dq \frac{Z_r(q)}{Z_r(q)^2 + a^2}. \quad (103)$$

As $a \rightarrow 0$, the logarithm comes only from the two endpoints. Excise a small distance ϵ from both endpoints. In the endpoint regions, $Z_r \simeq 2s_r p$, so the two endpoint contributions give

$$2 \int_0^\epsilon dp \frac{2s_r p}{(2s_r p)^2 + a^2} = \frac{1}{s_r} \log \frac{2s_r \epsilon}{a} + O(a^2). \quad (104)$$

On the remaining interval one may set $a = 0$. Using

$$\frac{1}{Z_r(q)} = \frac{1}{2} \left[\frac{1}{\cos q - c_r} + \frac{1}{r + 1 - \cos q} \right],$$

the finite bulk part is

$$\int_{-q_r+\epsilon}^{q_r-\epsilon} \frac{dq}{Z_r(q)} = \frac{1}{s_r} \log \frac{2s_r}{\epsilon} + \frac{J_r}{2} + O(\epsilon), \quad (105)$$

where J_r was defined in (98). Adding (104) and (105) cancels the auxiliary cutoff ϵ and gives

$$I_r(a) = \frac{1}{s_r} \log \frac{4s_r^2}{a} + \frac{J_r}{2} + o(1) = \frac{1}{s_r} \log \frac{16s_r^2 \exp(s_r J_r/2)}{\Gamma|\Omega|} + o(1). \quad (106)$$

Thus, in the endpoint convention used above,

$$C_r^{\text{end}} = 16s_r^2 \exp\left(\frac{s_r J_r}{2}\right). \quad (107)$$

This is not a one-parameter fit to the full lattice bubble. It is the matching constant of the reduced endpoint kernel after the regular terms in (99) are kept separately.

The expansion (101) is a small-external- argument expansion at fixed Γ . Its actual control parameters are

$$\Gamma|\Omega - c_r k|, \quad \Gamma|k|,$$

not only $|\Omega|$ and $|k|$. Thus the quadratic/logarithmic coefficients above describe the regime $\Gamma Q \ll 1$, where Q denotes a typical external scale. If Γ is taken large first at fixed nonzero (Ω, k) , the endpoint kernel crosses over to a different $O(\Gamma^{-2})$ form. This noncommutativity of limits is derived in Appendix A2. For the full retarded-advanced Green's function, the endpoint saturation is supplemented by smooth analytic terms from the nonsingular components; the resulting complex kernel is given in (A53).

The same constant is obtained at nonzero k . Define

$$\delta \equiv \Omega - c_r k, \quad D(q) \equiv \Omega - k \cos q = \delta - k(\cos q - c_r). \quad (108)$$

The finite- Γ endpoint kernel can be written as

$$\Delta_{\text{end}}(\Omega, k) = - \int_{-q_r}^{q_r} \frac{dq}{2\pi} \frac{Z_r(q) D(q)^2}{4Z_r(q)^2 + \Gamma^2 D(q)^2/4}. \quad (109)$$

Near either endpoint, $Z_r \simeq 2s_r p$ and $\cos q - c_r = O(p)$, so $D(q) = \delta + O(kp)$. The logarithmic piece is therefore obtained by replacing $D(q)$ by δ inside the singular endpoint region. Equivalently, split

$$D(q)^2 = \delta^2 + [D(q)^2 - \delta^2].$$

The first term gives the same integral $I_r(a)$, with $a = \Gamma|\delta|/4$, and hence

$$-\frac{\delta^2}{8\pi} I_r\left(\frac{\Gamma|\delta|}{4}\right) = -\frac{\delta^2}{8\pi s_r} \log \frac{C_r^{\text{end}}}{\Gamma|\delta|} + o(Q^2). \quad (110)$$

The second term is regular, since

$$D(q)^2 - \delta^2 = (\cos q - c_r) [-2\Omega k + k^2(\cos q + c_r)], \quad (111)$$

and the factor $\cos q - c_r$ cancels the endpoint zero in $Z_r(q)$. Thus the nonzero- k computation gives the same C_r^{end} once $\Delta_{r,\text{reg}}^{(-)}$ is kept separately. The only exception is the tangent

direction $\delta = \Omega - c_r k = 0$: there the leading endpoint mismatch vanishes, the $O(kp)$ part of $D(q)$ regulates the endpoint, and one should use the unexpanded kernel (109) rather than extracting a matching constant from the logarithmic formula alone.

Equivalently, if one extracts effective coefficients on a fixed small fitting window,

$$\Delta_r(0, k) \simeq -\hat{A}_k(r, \Gamma) k^2, \quad (112)$$

$$\Delta_r(\Omega, 0) \simeq -\hat{A}_\Omega(r, \Gamma) \Omega^2, \quad (113)$$

$$\Delta_{r,\text{odd-odd}}(\Omega, k) \simeq \hat{A}_{\Omega k}(r, \Gamma) \Omega k, \quad (114)$$

then the logarithmic Γ -drifts are

$$\frac{\partial \hat{A}_k}{\partial \log \Gamma} = -L_k(r), \quad L_k(r) = \frac{c_r^2}{8\pi s_r}, \quad (115)$$

$$\frac{\partial \hat{A}_\Omega}{\partial \log \Gamma} = -L_\Omega(r), \quad L_\Omega(r) = \frac{1}{8\pi s_r}, \quad (116)$$

$$\frac{\partial \hat{A}_{\Omega k}}{\partial \log \Gamma} = -L_{\Omega k}(r), \quad L_{\Omega k}(r) = \frac{c_r}{4\pi s_r}. \quad (117)$$

The constants left after adding $L_i(r) \log \Gamma$ are not universal: they depend on the fitting window and on regular parts of the full Green's function. The slopes (115)–(117), however, are fixed by the edge-bulk endpoint.

The $r = 1$ point is special because $c_r = 0$. Then

$$L_k(1) = 0, \quad L_{\Omega k}(1) = 0, \quad L_\Omega(1) = \frac{1}{8\pi}, \quad (118)$$

and the finite endpoint-regular integrals reduce to

$$A_k^{\text{reg}}(1) = \frac{1}{8\pi} \int_{-\pi/2}^{\pi/2} \frac{\cos q}{2 - \cos q} dq = \frac{1}{3\sqrt{3}} - \frac{1}{8}, \quad (119)$$

$$B_{\Omega k}^{\text{reg}}(1) = \frac{1}{4\pi} \int_{-\pi/2}^{\pi/2} \frac{dq}{2 - \cos q} = \frac{1}{3\sqrt{3}}. \quad (120)$$

Thus the previously used constants are recovered only at the symmetric value $r = 1$. For a generic value of r , the leading singular kernel is instead organized by the single endpoint combination $\Omega - (r - 1)k$.

A. Finite- Γ infrared cutoffs

There are two distinct infrared cutoffs at finite Γ . The first one is the endpoint boundary layer in the reduced edge-bulk bubble. Write

$$\delta \equiv \Omega - c_r k.$$

The logarithm in (95) comes from the interval where the distance p from an edge-bulk touching point obeys

$$p_{\min} \ll p \ll 1, \quad p_{\min} \sim \frac{\Gamma|\delta|}{2s_r},$$

up to order-one constants. The on-shell residue approximation $\Sigma_R(\omega, q) \simeq Z_r(q)/(\omega - \sin q + i0)$ is itself uniform only outside the narrower endpoint strip $p \lesssim \Gamma^{-1}$. Thus the clean logarithmic matching window is

$$\frac{1}{\Gamma^2} \ll |\delta| \ll \frac{1}{\Gamma}. \quad (121)$$

In this window $p_{\min} \gg \Gamma^{-1}$, and the singular term is controlled by

$$-\frac{\delta^2}{8\pi s_r} \log \frac{C_r}{\Gamma|\delta|}.$$

If instead

$$|\delta| \ll \Gamma^{-2}, \quad (122)$$

then the formal cutoff $p_{\min}$ lies inside the strip where the edge pole is already too close to the bulk branch point. In that ultra-small- δ regime one should not extract the matching constant C_r from the on-shell-residue formula. The logarithm is cut off by the branch-point strip at $p \sim \Gamma^{-1}$, so the remaining δ -dependence is analytic to this order, with a coefficient containing $\log \Gamma$ rather than $\log[1/(\Gamma|\delta|)]$. A direct full-Green probe at $r = 1$ and $\Gamma = 20$ indeed shows that the extracted $\log C$ drifts once $\Gamma^2|\delta| = O(1)$, but this is a delicate boundary-layer effect rather than a robust way to determine C_r .

The second cutoff is the residual static bubble if the exact finite- Γ local saddle normalization is replaced by its asymptotic value. Define

$$m_\Gamma(r) \equiv 2 - \Gamma \Pi_{RA}(0, 0; r, \Gamma). \quad (123)$$

The exact quadratic NLSM kernel uses $\Pi_{RA}(\Omega, k) - \Pi_{RA}(0, 0)$, so this mass is absent when the finite Γ saddle cancellation is kept exactly. If one instead substitutes $\Gamma \Pi_{RA}(0, 0) \rightarrow 2$,

the diagonal kernel contains the residual local piece

$$\Gamma \left[\Pi_{RA}(\Omega, k) - \frac{2}{\Gamma} \right] = -m_\Gamma(r) + \Gamma [\Pi_{RA}(\Omega, k) - \Pi_{RA}(0, 0)]. \quad (124)$$

The scaling of m_Γ is

$$m_\Gamma(r) = \frac{\mu_r}{\Gamma^2} + O(\Gamma^{-3}), \quad \mu_r = O(1). \quad (125)$$

This follows from the spectral-leakage identity derived in (A10): the defect of the monitored spectral sum rule is supported only on the finite bulk continuum, where $G_R = O(\Gamma^{-1})$.

Numerically, for $r = 1$, evaluating the leakage integral gives

Γ	20	40	80	160	320
$\Gamma^2 m_\Gamma$	1.83	2.11	2.27	2.36	2.41

which is consistent with (125).

This residual mass produces a correlation scale by comparison with the static stiffness:

$$A_k(r)k^2 \sim m_\Gamma(r) \quad \Rightarrow \quad k_m \sim \frac{1}{\Gamma} \sqrt{\frac{\mu_r}{A_k(r)}}.$$

Thus, if m_Γ is physically retained, the mass cutoff enters at $Q \sim \Gamma^{-1}$, whereas the ultra-small endpoint boundary layer begins only at $|\delta| \sim \Gamma^{-2}$. The mass therefore cuts off the long-wavelength physics parametrically earlier. The regime $|\delta| \ll \Gamma^{-2}$ is physically relevant only for an exactly massless kernel in which the finite- Γ local term has been subtracted, or tuned, with the same accuracy as $\Pi_{RA}(0, 0)$.

B. Numerical checks at representative r

We checked the logarithmic slopes using the existing direct quadrature code. For the k and Ω line cuts we used `src/collect_small_argument_direct_lines.jl` with

$$N_\omega = 8192, \quad N_q = 2048, \quad \omega_{\max} = 640,$$

on the windows

$$5 \times 10^{-4} \leq k \leq 2.5 \times 10^{-3}, \quad 5 \times 10^{-4} \leq \Omega \leq 1.5 \times 10^{-3}.$$

For the mixed term we used `src/estimate_edge_Aomegak.jl` at $\Omega = k = 10^{-3}$ with the same grid. For $r = 0.5$ and $r = 1.5$,

$$s_r = \frac{\sqrt{3}}{2}, \quad L_k = 0.0114860, \quad L_\Omega = 0.0459441, \quad L_{\Omega k} = \mp 0.0459441,$$

with the minus sign for $r = 0.5$ and the plus sign for $r = 1.5$.

r	Γ	$\hat{A}_k + L_k \log \Gamma$	$\hat{A}_\Omega + L_\Omega \log \Gamma$	$\hat{A}_{\Omega k} + L_{\Omega k} \log \Gamma$
0.5	40	0.13754	0.51452	-0.14710
0.5	80	0.12969	0.49590	-0.14300
1.5	40	0.17189	0.44960	0.56720
1.5	80	0.17021	0.45363	0.56814

The shifted combinations are approximately constant between $\Gamma = 40$ and 80, while the unshifted fitted coefficients drift with the sign and magnitude predicted by (115)–(117). For example, at $r = 1.5$ the raw line-cut coefficients change as

$$\hat{A}_k : 0.12952 \rightarrow 0.11988, \quad \hat{A}_\Omega : 0.28012 \rightarrow 0.25230, \quad \hat{A}_{\Omega k} : 0.39772 \rightarrow 0.36681,$$

when Γ is doubled from 40 to 80, consistent with a negative $\log \Gamma$ drift. The $r = 0.5$ mixed coefficient has the opposite logarithmic slope because $L_{\Omega k} < 0$, and the shifted values show the same stabilization. These checks support the generic- r endpoint prediction while preserving the earlier $r = 1$ constants as a special case.

We also verified the regular coefficients (100) directly in the reduced endpoint model using `src/verify_generic_r_regular_terms.jl`. The numerical test integrates the finite- Γ endpoint kernel

$$\Delta_{\text{end}}(\Omega, k) = \int_{-q_r}^{q_r} \frac{dq}{2\pi} \frac{-Z_r(q)(\Omega - k \cos q)^2}{4Z_r(q)^2 + \Gamma^2(\Omega - k \cos q)^2/4},$$

subtracts the logarithmic term in (95), fixes the one remaining $(\Omega - c_r k)^2$ constant from the Ω -line, and then extracts A_k^{reg} and $B_{\Omega k}^{\text{reg}}$. At $\Gamma = 80$ the comparison is

r	$A_{k,\text{num}}^{\text{reg}}$	A_k^{reg}	$B_{\Omega k,\text{num}}^{\text{reg}}$	$B_{\Omega k}^{\text{reg}}$
0.5	0.02197	0.02097	0.37215	0.37527
1.0	0.06734	0.06745	0.19028	0.19245
1.5	0.06570	0.06727	0.09689	0.10040

The agreement is at the few-percent level on the finite fitting window used for the check. The full Green’s-function quadrature verifies the logarithmic slopes cleanly; its absolute regular constants also contain nonsingular lattice contributions, so the reduced endpoint table is the sharper test of the analytic regular terms in (100).

The analytical constant C_r^{end} in (107) is only the reduced endpoint matching constant. It is not, by itself, a verification of the full lattice Green's function. The full Green's function contains nonsingular regular terms that are also $O(\Omega^2)$ on the $k = 0$ line, so the constant multiplying the logarithm should be extracted from the numerical full bubble after the logarithmic slope has been fixed.

For this purpose it is better to work with the logarithm of the constant. On the $k = 0$ line define

$$\ell_r^{\text{full}}(\Omega; \Gamma) \equiv \log C_{r,\text{est}}^{\text{full}} = \log(\Gamma|\Omega|) - \frac{\Delta_{\text{full}}(\Omega, 0)}{L_{\Omega}(r)\Omega^2}, \quad L_{\Omega}(r) = \frac{1}{8\pi s_r}, \quad (126)$$

where

$$\Delta_{\text{full}}(\Omega, 0) = \Gamma \text{Re} \left[\frac{\Pi_{RA}(\Omega, 0) + \Pi_{RA}(-\Omega, 0)}{2} - \Pi_{RA}(0, 0) \right]$$

is computed from the full orbital Green's function. In this convention all regular $O(\Omega^2)$ terms of the full bubble are absorbed into ℓ_r^{full} . Thus ℓ_r^{full} should be compared with $\log C_r^{\text{end}}$, not with C_r^{end} itself.

This extraction is numerically more delicate than the logarithmic-slope check, because it divides a small $O(\Omega^2)$ difference by Ω^2 . In particular, the frequency mesh must resolve the external frequency itself. A coarse dense-region spacing $d\omega = 1/(20\Gamma)$ gives an artificially low ℓ_r^{full} near $r = 1$. We therefore use a refined spacing $d\omega = 1/(100\Gamma)$ for the quoted full-Green values.

We performed this extraction with `src/estimate_full_Cr.jl`. The script evaluates the full orbital trace and, in the same run, the σ_x $--$ component. The raw complex bubbles and grids are saved in `data/full_green_lnCr_refined.jld2`. The table uses $\Gamma = 80$, $\Omega = 1.5 \times 10^{-3}$, and hence $\Gamma\Omega = 0.12$:

r	$\log C_r^{\text{end}}$	$\ell_r^{\text{full}}(1.5 \times 10^{-3})$	$\ell_r^{--}(1.5 \times 10^{-3})$	$\ell_r^{\text{full}} - \log C_r^{\text{end}}$
0.5	4.52692	4.52236	4.51515	-0.00456
1.0	3.98179	3.95438	3.94670	-0.02740
1.5	3.03124	3.00945	3.00178	-0.02179

The full result is indeed dominated by the σ_x $--$ bubble: the middle columns differ only at the 10^{-2} level in $\log C$. The residual difference from $\log C_r^{\text{end}}$ is much smaller than what one finds on the coarse frequency mesh, and is comparable to remaining finite- $\Gamma\Omega$ and finite-mesh uncertainties.

For reference, at the most sensitive point $r = 1$ the frequency convergence is

$d\omega$	N_ω	$\ell_1^{\text{full}}(1.5 \times 10^{-3})$	$\ell_1^{--}(1.5 \times 10^{-3})$
$1/(20\Gamma)$	8952	3.86625	3.85870
$1/(50\Gamma)$	18552	3.94260	3.93495
$1/(100\Gamma)$	34552	3.95438	3.94670
$1/(200\Gamma)$	66552	3.95900	3.95134

This convergence data is saved in `data/full_green_lnCr_r1_frequency_convergence.jld2`. The main lesson is that the full integral is dominated by the $--$ bubble, but the matching constant is not a robust quantity unless the regular $O(\Omega^2)$ terms and the numerical frequency resolution are controlled. The logarithmic slopes are much more reliable than the absolute constant.

VII. Trivial- $r > 2$ Bubble

For $r > 2$, the boundary is in the trivial phase. The condition $|r - \cos q| < 1$ has no solution, so the edge-pole interval $\mathcal{I}_r$ is empty. The endpoint mechanism responsible for $\Omega^2 \log |\Omega|$, $k^2 \log \Gamma$, and $\Omega k \log \Gamma$ is therefore absent. The full retarded-advanced bubble is analytic at small (Ω, k) .

It is convenient to fit the unsymmetrized full bubble as

$$\Gamma \Pi_{RA}(\Omega, k) = C_0 + L_\Omega \Omega + L_k k + Q_{\Omega\Omega} \Omega^2 + Q_{\Omega k} \Omega k + Q_{kk} k^2 + \dots \quad (127)$$

The large- Γ expansion is controlled by the broadening-dominated Green's function. Since the trivial bulk is gapped, the self-energy is analytic near zero frequency and obeys $\Sigma_R(z, q) = 1/z + O(z^{-2})$ at large $|z|$. For $\omega = \Gamma x/2$,

$$G_R\left(\frac{\Gamma x}{2} + \Omega, q + k\right) = \frac{2}{\Gamma} \frac{1}{x + i + 2\Omega/\Gamma} + \frac{4}{\Gamma^2} \frac{h_r(q + k)}{(x + i)^2} + O(\Gamma^{-3}), \quad (128)$$

where

$$h_r(q) = \sin q \sigma_x + (r - \cos q) \sigma_z.$$

The scalar part gives

$$\Gamma \Pi_{RA}^{(0)}(\Omega, k) = \frac{2}{1 - i\Omega/\Gamma} = 2 + \frac{2i\Omega}{\Gamma} - \frac{2\Omega^2}{\Gamma^2} + O(\Gamma^{-3}Q^3). \quad (129)$$

The first momentum-dependent term comes from $h_r(q+k)h_r(q)$. Subtracting the $k=0$ value,

$$\begin{aligned}\Gamma [\Pi_{RA}(0, k) - \Pi_{RA}(0, 0)]_{h^2} &= \frac{2}{\Gamma^2} \int_{-\pi}^{\pi} \frac{dq}{2\pi} \text{tr} [h_r(q+k)h_r(q) - h_r(q)^2] \\ &= \frac{4(\cos k - 1)}{\Gamma^2} = -\frac{2k^2}{\Gamma^2} + O(k^4/\Gamma^2).\end{aligned}\quad (130)$$

The trace of h_r vanishes and the scalar denominator is independent of q , so there is no linear k term and no Ωk term at this order.

The constant term is most cleanly obtained from the spectral-leakage identity (A10). In the trivial phase the continuum exhausts the self-energy spectral weight:

$$\int \frac{d\omega}{2\pi} [-2 \text{Im} \Sigma_R(\omega, q)] = 1$$

for each q . Since $\text{tr} [G_R P_- G_A] = 4/\Gamma^2 + O(\Gamma^{-3})$ on the finite bulk continuum,

$$C_0(r, \Gamma) = \Gamma \Pi_{RA}(0, 0) = 2 - \frac{4}{\Gamma^2} + O(\Gamma^{-3}), \quad r > 2. \quad (131)$$

Combining these pieces gives the leading trivial-phase coefficients

$$\begin{aligned}C_0 &= 2 - \frac{4}{\Gamma^2} + O(\Gamma^{-3}), & L_\Omega &= \frac{2i}{\Gamma} + O(\Gamma^{-3}), & L_k &= O(\Gamma^{-3}), \\ Q_{\Omega\Omega} &= -\frac{2}{\Gamma^2} + O(\Gamma^{-3}), & Q_{kk} &= -\frac{2}{\Gamma^2} + O(\Gamma^{-3}), & Q_{\Omega k} &= O(\Gamma^{-3}).\end{aligned}\quad (132)$$

The leading coefficients are independent of r as long as $r > 2$. The r -dependence first enters through subleading corrections to the analytic large- Γ expansion.

We checked these predictions at $r=3$ by direct full-Green quadrature using `src/analyze_trivial_phase`. The raw bubbles and fit outputs are saved in `data/trivial_r3_bubble_scan.jld2`. The table shows the scaled coefficients; the first four columns should approach 4, 2, -2, -2, while the last two columns remain $O(1)$ because the corresponding coefficients start only at $O(\Gamma^{-3})$.

Γ	$\Gamma^2 m_{\text{leak}}$	$\Gamma \text{Im} L_\Omega$	$\Gamma^2 \text{Re} Q_{\Omega\Omega}$	$\Gamma^2 \text{Re} Q_{kk}$	$\Gamma^3 \text{Im} L_k$	$\Gamma^3 \text{Re} Q_{\Omega k}$
20	2.95290	2.00183	-3.12290	-2.63531	-3.97411	15.15975
40	3.50134	2.00078	-2.65491	-2.47261	-5.46633	21.94952
80	3.77522	2.00011	-2.26667	-2.28032	-6.22272	24.90956
160	3.89637	1.99988	-2.03910	-2.15607	-6.56212	25.80893

Here m_{leak} denotes the positive residual $2 - \Gamma \Pi_{RA}(0, 0)$ evaluated through (A10), which is more stable than subtracting the tail-corrected constant at the largest Γ . The numerical trend confirms that the trivial phase has no separate $\Gamma Q \ll 1$ endpoint regime. After the constant piece is subtracted, the small- (Ω, k) kernel decays algebraically with Γ , with the leading terms shown in (132).

VIII. One-loop RG of the replica sigma model

We now use the replica-sector kernel to identify the sigma-model stiffness and to derive its one-loop RG flow. Since the full KK bubble is smooth and strongly suppressed at large Γ , we keep only the retarded-advanced part in the replica sector. From (82), (83), and (80), the Euclidean quadratic action of the traceless replica mode is

$$S_E^{(2)}[Y] = \frac{\Gamma}{4} \int_Q \left[A_k k^2 - A_{\Omega k} \Omega k + \frac{\Omega^2}{8\pi} \log \frac{C_\Omega}{\Gamma |\Omega|} + \dots \right] \text{Tr}_R[Y(-Q)Y(Q)]. \quad (133)$$

Matching the spatial-gradient term to the standard normalization $\frac{\lambda}{8} \int_Q k^2 \text{Tr}_R[Y(-Q)Y(Q)]$, we identify

$$\frac{\lambda}{8} \equiv \frac{\Gamma A_k}{4}, \quad \lambda = 2\Gamma A_k. \quad (134)$$

Using the large- Γ value of A_k ,

$$\lambda \xrightarrow{\Gamma \rightarrow \infty} \Gamma \left(\frac{2}{3\sqrt{3}} - \frac{1}{4} \right) = \Gamma \frac{8 - 3\sqrt{3}}{12\sqrt{3}}. \quad (135)$$

It is also convenient to define

$$v \equiv \frac{A_{\Omega k}}{A_k}, \quad \kappa \equiv \frac{1}{8\pi A_k}, \quad \rho(\Omega) \equiv \kappa \log \frac{C_\Omega}{\Gamma |\Omega|}. \quad (136)$$

Then (133) becomes

$$S_E^{(2)}[Y] = \frac{\lambda}{8} \int_Q [k^2 - v \Omega k + \rho(\Omega) \Omega^2 + \dots] \text{Tr}_R[Y(-Q)Y(Q)]. \quad (137)$$

The mixed term can be removed by the shear

$$\tilde{k} = k - \frac{v}{2} \Omega, \quad (138)$$

under which

$$k^2 - v \Omega k + \rho(\Omega) \Omega^2 = \tilde{k}^2 + \tilde{\rho}(\Omega) \Omega^2, \quad \tilde{\rho}(\Omega) \equiv \rho(\Omega) - \frac{v^2}{4}. \quad (139)$$

More generally, the omitted analytic Ω^2 terms in (133) can be absorbed into a regular constant r , so the time kernel takes the form

$$\tilde{\rho}(\Omega) = r + \kappa \log \frac{\Omega_0}{|\Omega|}, \quad \Omega_0 \equiv \frac{C_\Omega}{\Gamma}. \quad (140)$$

At asymptotically small $|\Omega|$, the logarithm wins over the regular part, so $\tilde{\rho}(\Omega)$ remains positive. We therefore perform the RG in the sheared frame and drop the tilde on k from now on.

The nonlinear completion of the replica sector is obtained by introducing

$$X(\tau, x) \in SU(R), \quad J_\mu \equiv X^{-1} \partial_\mu X, \quad (141)$$

so that $J_\mu = i\partial_\mu Y + O(Y^2)$. The replica sigma model is then

$$S_E[X] = \frac{\lambda}{8} \left[\int d\tau dx \operatorname{Tr}_R(J_x^2) + \int_Q \rho(\Omega) \operatorname{Tr}_R(J_\tau(-Q)J_\tau(Q)) \right]. \quad (142)$$

The logarithm therefore remains attached to the time-derivative current bilinear; it is not replaced by a local $(\partial_\tau X)^2$ term.

To carry out Wilsonian RG, split

$$X = X_s X_f, \quad X_f = e^{iY_f}, \quad (143)$$

where Y_f contains only shell modes. Writing

$$J_\mu^{(s)} \equiv X_s^{-1} \partial_\mu X_s, \quad (144)$$

we have

$$J_\mu = X_f^{-1} J_\mu^{(s)} X_f + X_f^{-1} \partial_\mu X_f. \quad (145)$$

Expanding $X_f = e^{iY_f}$ gives

$$J_\mu = J_\mu^{(s)} + i\nabla_\mu Y_f - \frac{1}{2}[Y_f, \nabla_\mu Y_f] - \frac{1}{2}[Y_f, [Y_f, J_\mu^{(s)}]] + O(Y_f^3), \quad (146)$$

with the background covariant derivative

$$\nabla_\mu Y_f \equiv \partial_\mu Y_f + i[J_\mu^{(s)}, Y_f]. \quad (147)$$

Substituting (146) into (142), the linear term in Y_f vanishes after integrating by parts. The quadratic fast action is

$$S_f[Y_f] = \frac{\lambda}{8} \int_Q \mathcal{K}(Q) \operatorname{Tr}_R[Y_f(-Q)Y_f(Q)], \quad \mathcal{K}(Q) = k^2 + \rho(\Omega)\Omega^2. \quad (148)$$

The logarithm must therefore be kept inside the fast-mode propagator throughout the shell integration.

Choose a basis T^a of $su(R)$ with

$$\text{Tr}_R(T^a T^b) = \frac{1}{2} \delta^{ab}, \quad Y_f = Y_f^a T^a. \quad (149)$$

Then

$$\langle Y_f^a(Q) Y_f^b(-Q) \rangle_> = \frac{8}{\lambda} \delta^{ab} \frac{1}{\mathcal{K}(Q)}, \quad (150)$$

because

$$\text{Tr}_R[Y_f(-Q) Y_f(Q)] = \frac{1}{2} Y_f^a(-Q) Y_f^a(Q). \quad (151)$$

This factor of $1/2$ from the trace normalization is the source of the missing factor of two in the naive estimate. and the shell average at coincident points is

$$I_> \equiv \int_> \frac{d\Omega dk}{(2\pi)^2} \frac{1}{\mathcal{K}(Q)}. \quad (152)$$

For an infinitesimal Wilson shell defined by $\Lambda e^{-d\ell} < \sqrt{\mathcal{K}(Q)} < \Lambda$, the logarithm varies only slowly across the shell. Freezing it at the running shell value

$$\rho_\ell \equiv r + \kappa\ell, \quad (153)$$

with $|\Omega| \sim \Omega_0 e^{-\ell}$, we obtain

$$I_> \simeq \int_> \frac{d\Omega dk}{(2\pi)^2} \frac{1}{k^2 + \rho_\ell \Omega^2} = \frac{d\ell}{2\pi\sqrt{\rho_\ell}} + O(d\ell^2). \quad (154)$$

So the logarithmic time kernel does affect the RG: it reduces the one-loop shell weight by the factor $1/\sqrt{\rho_\ell}$, which slowly decreases with scale.

The terms quadratic in both $J_\mu^{(s)}$ and Y_f renormalize the slow action. Using

$$\sum_a [T^a, [T^a, M]] = R M \quad \text{for } M \in su(R), \quad (155)$$

one finds

$$S_E^{\text{eff}}[X_s] = \frac{\lambda - 2RI_>}{8} \left[\int d\tau dx \text{Tr}_R((J_x^{(s)})^2) + \int_Q \rho(\Omega) \text{Tr}_R(J_\tau^{(s)}(-Q) J_\tau^{(s)}(Q)) \right]. \quad (156)$$

Thus the spatial-gradient term and the logarithmic time-derivative term are renormalized by the same loop factor. However, this does *not* mean that the logarithmic kernel is inert. It

affects the RG in two ways. First, it enters the shell integral through (154). Second, under the rescaling $\Omega \rightarrow e^{d\ell}\Omega$,

$$\rho(\Omega e^{-d\ell}) = \rho(\Omega) + \kappa d\ell, \quad (157)$$

so the regular Ω^2 coefficient shifts at tree level. Equivalently, if $\rho(\Omega) = r + \kappa \log(\Omega_0/|\Omega|)$, then

$$\frac{d\kappa}{d\ell} = 0 + O(\lambda^2), \quad \frac{dr}{d\ell} = \kappa + O(\lambda). \quad (158)$$

At this order the loop correction does not generate a separate beta function for κ ; it only renormalizes the overall stiffness.

Reading off the renormalized coupling from (156),

$$\lambda(\ell + d\ell) = \lambda(\ell) - 2R I_>, \quad (159)$$

and using (154), we obtain

$$\frac{d\lambda}{d\ell} = -\frac{R}{\pi\sqrt{\rho_\ell}} + O(\lambda^{-1}). \quad (160)$$

Thus the replica sector has the same one-loop algebraic structure as the principal chiral model, but with a scale-dependent shell weight set by the logarithmic time kernel. In the replica continuation relevant here, $R \rightarrow 1$, the one-loop flow is

$$\beta_\lambda = -\frac{1}{\pi\sqrt{\rho_\ell}} + O(\lambda^{-1}), \quad (161)$$

while $\rho_\ell = r + \kappa\ell$ grows logarithmically with the RG scale. Consequently the logarithmic time kernel suppresses the growth of λ at long distances through the factor $1/\sqrt{\rho_\ell}$. In the present convention, λ is a true stiffness, so the negative sign of (160) means that the stiffness decreases under coarse graining.

The one-loop flow can be integrated explicitly. For general R ,

$$\lambda(\ell) = \lambda_0 - \frac{2R}{\pi\kappa} \left(\sqrt{r + \kappa\ell} - \sqrt{r} \right), \quad \kappa > 0. \quad (162)$$

In the limit $\kappa \rightarrow 0$, this reduces smoothly to the isotropic result

$$\lambda(\ell) = \lambda_0 - \frac{R}{\pi\sqrt{r}} \ell. \quad (163)$$

For the standard isotropic normalization $r = 1$, one recovers

$$\frac{d\lambda}{d\ell} = -\frac{R}{\pi}, \quad \lambda(\ell) = \lambda_0 - \frac{R}{\pi} \ell. \quad (164)$$

The scale ℓ_* at which the stiffness is driven to zero, $\lambda(\ell_*) = 0$, estimates the onset of the disordered regime. Without the logarithmic term,

$$\ell_*^{(0)} = \frac{\pi\sqrt{r}}{R} \lambda_0 \quad \left(\ell_*^{(0)} = \frac{\pi}{R} \lambda_0 \text{ for } r = 1 \right). \quad (165)$$

With the logarithmic kernel present,

$$\ell_* = \frac{\pi\sqrt{r}}{R} \lambda_0 + \frac{\pi^2\kappa}{4R^2} \lambda_0^2. \quad (166)$$

Thus the logarithmic term slows the flow toward disorder. Instead of a constant velocity $d\lambda/d\ell = -R/\pi$, the magnitude of the beta function decays as

$$|\beta_\lambda| \sim \frac{R}{\pi\sqrt{\kappa\ell}} \quad (\ell \gg r/\kappa), \quad (167)$$

so the stiffness decreases only as $\lambda(\ell) \sim \lambda_0 - \text{const} \times \sqrt{\ell}$. Equivalently, the disordering scale is parametrically larger:

$$\ell_* - \ell_*^{(0)} = \frac{\pi^2\kappa}{4R^2} \lambda_0^2. \quad (168)$$

If one interprets $\xi \sim e^{\ell_*}$ as the correlation length, then the correlation length is

$$\xi \sim \xi_0 e^{\ell_*} = \xi_0 \exp \left[\frac{\pi\sqrt{r}}{R} \lambda_0 + \frac{\pi^2\kappa}{4R^2} \lambda_0^2 \right], \quad (169)$$

to be compared with the isotropic result without the logarithmic kernel,

$$\xi^{(0)} \sim \xi_0 \exp \left[\frac{\pi\sqrt{r}}{R} \lambda_0 \right]. \quad (170)$$

Thus

$$\frac{\xi}{\xi^{(0)}} = \exp \left[\frac{\pi^2\kappa}{4R^2} \lambda_0^2 \right]. \quad (171)$$

The logarithmic Ω^2 term therefore produces a parametrically larger correlation length. For moderate bare stiffness, where $\kappa\lambda_0 \ll 4R\sqrt{r}/\pi$, the extra term in the exponent is a small correction. For sufficiently large λ_0 , however, the quadratic term dominates, and the correlation length crosses over from the usual $\exp(\text{const} \times \lambda_0)$ behavior to the stronger form $\exp(\text{const} \times \lambda_0^2)$.

Using the microscopic large- Γ expressions

$$\lambda_0 \simeq 2\Gamma A_k, \quad \kappa = \frac{1}{8\pi A_k}, \quad (172)$$

and taking $R \rightarrow 1$, one obtains

$$\ell_* \simeq 2\pi A_k \sqrt{r} \Gamma + \frac{\pi A_k}{8} \Gamma^2, \quad (173)$$

so that at large Γ

$$\xi \sim \xi_0 \exp \left[2\pi A_k \sqrt{r} \Gamma + \frac{\pi A_k}{8} \Gamma^2 \right]. \quad (174)$$

At the one-loop level, this is the main physical consequence of the logarithmic time kernel: it does not stop the flow toward disorder, but it slows that flow enough to promote the correlation length from the usual exponential dependence to a super-exponential one in the microscopic stiffness.

IX. Generic- r One-loop RG in the Ballistic Frame

The RG discussion above was written in the symmetric case $r = 1$, where the nonanalytic piece sits on Ω^2 . For a generic topological value $0 < r < 2$, Section VI showed that the singular bubble depends instead on the endpoint combination

$$\delta_r \equiv \Omega - c_r k, \quad c_r = r - 1, \quad s_r = \sqrt{r(2-r)}. \quad (175)$$

Therefore the natural low-energy frame is not the laboratory frame (Ω, k) , but the ballistic frame aligned with the edge velocity c_r .

To avoid clashing with the microscopic mass parameter r , in this section we denote the regular analytic coefficient of the time kernel by $\rho_{0,r}$ rather than by r . The quadratic replica action consistent with (101) can be written as

$$S_{E,r}^{(2)}[Y] = \frac{\Gamma}{4} \int_Q \left[A_{k,r} k^2 - A_{\Omega k,r} \Omega k + A_{\Omega,r} \Omega^2 + \frac{(\Omega - c_r k)^2}{8\pi s_r} \log \frac{\Omega_{0,r}}{|\Omega - c_r k|} + \dots \right] \text{Tr}_R[Y(-Q)Y(Q)]. \quad (176)$$

Here $\Omega_{0,r} \sim C_r/\Gamma$ is the matching scale, while $A_{k,r}$, $A_{\Omega k,r}$, and $A_{\Omega,r}$ are finite analytic coefficients. They depend on the convention used to separate the regular and logarithmic pieces. In the reduced endpoint convention of (100) and (107),

$$A_{\Omega,r}^{\text{end}} = 0, \quad A_{\Omega k,r}^{\text{end}} = B_{\Omega k}^{\text{reg}}(r) = \frac{J_r}{4\pi}, \quad A_{k,r}^{\text{end}} = A_k^{\text{reg}}(r) = \frac{rJ_r - q_r}{4\pi}. \quad (177)$$

The full lattice Green's function shifts these coefficients by additional regular $O(1)$ terms, but it does not change the universal logarithmic coefficient $1/(8\pi s_r)$.

The singular direction is made explicit by the unit-Jacobian change of variables

$$\varpi \equiv \Omega - c_r k, \quad p \equiv k. \quad (178)$$

In real space this is the comoving coordinate system

$$\tau' = \tau, \quad x' = x - c_r \tau, \quad (179)$$

so ϖ is conjugate to τ' , and

$$\partial_{\tau'} = \partial_\tau + c_r \partial_x, \quad \partial_{x'} = \partial_x. \quad (180)$$

Thus the logarithmic kernel is attached to the convective derivative along the edge trajectory rather than to ∂_τ alone.

Substituting $\Omega = \varpi + c_r p$ into (176) gives

$$S_{E,r}^{(2)}[Y] = \frac{\Gamma}{4} \int_Q \left[\bar{A}_{k,r} p^2 - \bar{A}_{\varpi p,r} \varpi p + \left(\bar{A}_{\varpi,r} + \frac{1}{8\pi s_r} \log \frac{\Omega_{0,r}}{|\varpi|} \right) \varpi^2 + \dots \right] \text{Tr}_R[Y(-Q)Y(Q)], \quad (181)$$

with

$$\bar{A}_{k,r} = A_{k,r} - c_r A_{\Omega k,r} + c_r^2 A_{\Omega,r}, \quad (182)$$

$$\bar{A}_{\varpi p,r} = A_{\Omega k,r} - 2c_r A_{\Omega,r}, \quad (183)$$

$$\bar{A}_{\varpi,r} = A_{\Omega,r}. \quad (184)$$

In the reduced endpoint convention (177), these become

$$\bar{A}_{k,r}^{\text{end}} = A_k^{\text{reg}}(r) - c_r B_{\Omega k}^{\text{reg}}(r) = \frac{J_r - q_r}{4\pi}, \quad \bar{A}_{\varpi p,r}^{\text{end}} = \frac{J_r}{4\pi}, \quad \bar{A}_{\varpi,r}^{\text{end}} = 0. \quad (185)$$

Since $J_r - q_r > 0$ for $0 < r < 2$, the ballistic spatial stiffness is positive throughout the topological phase.

Matching the p^2 term to the standard normalization gives

$$\frac{\lambda_r}{8} \equiv \frac{\Gamma \bar{A}_{k,r}}{4}, \quad \lambda_r = 2\Gamma \bar{A}_{k,r}. \quad (186)$$

It is then natural to define

$$v_r \equiv \frac{\bar{A}_{\varpi p,r}}{\bar{A}_{k,r}}, \quad \kappa_r \equiv \frac{1}{8\pi s_r \bar{A}_{k,r}}, \quad \rho_r(\varpi) \equiv \rho_{0,r} + \kappa_r \log \frac{\Omega_{0,r}}{|\varpi|}, \quad \rho_{0,r} \equiv \frac{\bar{A}_{\varpi,r}}{\bar{A}_{k,r}}. \quad (187)$$

Then (181) becomes

$$S_{E,r}^{(2)}[Y] = \frac{\lambda_r}{8} \int_Q [p^2 - v_r \varpi p + \rho_r(\varpi) \varpi^2 + \dots] \text{Tr}_R[Y(-Q)Y(Q)]. \quad (188)$$

In the reduced endpoint convention,

$$\lambda_r^{\text{end}} = \Gamma \frac{J_r - q_r}{2\pi}, \quad v_r^{\text{end}} = \frac{J_r}{J_r - q_r}, \quad \kappa_r^{\text{end}} = \frac{1}{2s_r(J_r - q_r)}. \quad (189)$$

At $r = 1$, where $c_r = 0$, $q_r = \pi/2$, and $J_r = 4\pi/(3\sqrt{3})$, these reduce to the $r = 1$ couplings used above.

As in (139), the remaining analytic mixed term is removed by the shear

$$\tilde{p} = p - \frac{v_r}{2} \varpi, \quad (190)$$

under which

$$p^2 - v_r \varpi p + \rho_r(\varpi) \varpi^2 = \tilde{p}^2 + \tilde{\rho}_r(\varpi) \varpi^2, \quad \tilde{\rho}_r(\varpi) \equiv \tilde{\rho}_{0,r} + \kappa_r \log \frac{\Omega_{0,r}}{|\varpi|}, \quad (191)$$

with

$$\tilde{\rho}_{0,r} \equiv \rho_{0,r} - \frac{v_r^2}{4}. \quad (192)$$

The omitted analytic $O(\varpi^2)$ terms from the full Green's function only shift $\tilde{\rho}_{0,r}$ by a finite amount. The logarithmic coefficient κ_r is fixed by the endpoint data. In the reduced endpoint convention, $\rho_{0,r}^{\text{end}} = 0$, so

$$\tilde{\rho}_{0,r}^{\text{end}} = -\frac{(v_r^{\text{end}})^2}{4} = -\frac{J_r^2}{4(J_r - q_r)^2}. \quad (193)$$

This negative $O(1)$ offset is convention-dependent: the full lattice Green's function adds further analytic ϖ^2 terms. The RG below is therefore understood to start at scales where the logarithm has already made $\tilde{\rho}_r(\varpi)$ positive.

The nonlinear completion is therefore the same principal-chiral replica sigma model as before, but written in the comoving coordinates (τ', x') :

$$S_{E,r}[X] = \frac{\lambda_r}{8} \left[\int d\tau' dx' \text{Tr}_R(J_{x'}^2) + \int_Q \tilde{\rho}_r(\varpi) \text{Tr}_R(J_{\tau'}(-Q)J_{\tau'}(Q)) \right], \quad (194)$$

where $J_{\mu'} \equiv X^{-1} \partial_{\mu'} X$. In other words, the generic $0 < r < 2$ theory is not isotropic in the laboratory variables, but it becomes the same logarithmic sigma model after passing to the ballistic frame selected by the edge velocity c_r .

Because the derivation of the fast-mode expansion depends only on the quadratic kernel, the one-loop calculation goes through unchanged with (Ω, k) replaced by $(\varpi, \tilde{p})$. The fast propagator is

$$\langle Y_f^a(Q) Y_f^b(-Q) \rangle_> = \frac{8}{\lambda_r} \delta^{ab} \frac{1}{\tilde{p}^2 + \tilde{\rho}_r(\varpi) \varpi^2}, \quad (195)$$

and the infinitesimal shell integral is

$$I_>^{(r)} \equiv \int_> \frac{d\varpi d\tilde{p}}{(2\pi)^2} \frac{1}{\tilde{p}^2 + \tilde{\rho}_r(\varpi) \varpi^2} \simeq \frac{d\ell}{2\pi \sqrt{\tilde{\rho}_{r,\ell}}}, \quad (196)$$

with

$$\tilde{\rho}_{r,\ell} = \tilde{\rho}_{0,r} + \kappa_r \ell, \quad |\varpi| \sim \Omega_{0,r} e^{-\ell}. \quad (197)$$

Therefore the couplings obey

$$\frac{d\kappa_r}{d\ell} = 0, \quad \frac{d\tilde{\rho}_{0,r}}{d\ell} = \kappa_r, \quad \frac{d\lambda_r}{d\ell} = -\frac{R}{\pi \sqrt{\tilde{\rho}_{r,\ell}}} + O(\lambda_r^{-1}). \quad (198)$$

Thus the one-loop beta function has exactly the same algebraic form as in the $r = 1$ analysis. The qualitative difference is purely kinematic: the shell is centered on the ballistic line $\Omega = c_r k$, not on $\Omega = 0$.

For any reference scale ℓ_0 at which $\tilde{\rho}_{r,\ell_0} > 0$, the flow integrates to

$$\lambda_r(\ell) = \lambda_r(\ell_0) - \frac{2R}{\pi \kappa_r} \left[\sqrt{\tilde{\rho}_{r,\ell}} - \sqrt{\tilde{\rho}_{r,\ell_0}} \right]. \quad (199)$$

So once the logarithm dominates the analytic offset, the stiffness again decreases only as $\lambda_r(\ell) \sim \lambda_r(\ell_0) - \text{const} \times \sqrt{\ell}$. The generic- r anisotropy therefore changes the bare stiffness, the effective velocity, and the direction of the soft mode, but it does not change the conclusion that the logarithmic kernel slows the flow toward disorder and enhances the correlation length parametrically.

A. Running of the aspect ratio

It is useful to separate three distinct notions of anisotropy. First, the ballistic velocity c_r selects the singular direction $\delta_r = \Omega - c_r k$. This is fixed by the endpoint kinematics of the fermion bubble and is not a coupling of the sigma model itself. Second, the analytic mixed term in the ballistic frame is encoded in the dimensionless ratio v_r . Because the

one-loop shell integration renormalizes the whole bracket in (188) by the same factor, v_r is not renormalized at this order:

$$\frac{dv_r}{d\ell} = 0 + O(\lambda_r^{-1}). \quad (200)$$

Thus the orientation of the soft ridge is fixed; in the original laboratory variables the shell remains centered on the ballistic line $\Omega = c_r k$.

The actual running anisotropy is the relative weight of $\tilde{p}^2$ and ϖ^2 after both shears have been performed. At shell scale ℓ , the quadratic kernel is

$$\tilde{p}^2 + \tilde{\rho}_{r,\ell} \varpi^2, \quad \tilde{\rho}_{r,\ell} = \tilde{\rho}_{0,r} + \kappa_r \ell.$$

If one defines the momentum-space shell aspect ratio by

$$a_r(\ell) \equiv \sqrt{\tilde{\rho}_{r,\ell}}, \quad (201)$$

then typical shell momenta satisfy

$$|\tilde{p}|_\ell \sim \Lambda e^{-\ell}, \quad |\varpi|_\ell \sim \frac{\Lambda e^{-\ell}}{a_r(\ell)}. \quad (202)$$

Equivalently, in real space the characteristic coarse-grained scales obey

$$\frac{L_{x'}}{L_{\tau'}} \sim a_r(\ell) = \sqrt{\tilde{\rho}_{r,\ell}}.$$

Therefore the aspect ratio does flow:

$$\frac{d \log a_r}{d\ell} = \frac{1}{2} \frac{d \log \tilde{\rho}_{r,\ell}}{d\ell} = \frac{\kappa_r}{2\tilde{\rho}_{r,\ell}}, \quad (203)$$

so at large scales

$$a_r(\ell) \sim \sqrt{\kappa_r \ell} \quad (\ell \gg |\tilde{\rho}_{0,r}|/\kappa_r). \quad (204)$$

The anisotropy therefore grows only logarithmically.

This logarithmic running is not the same as flowing to a new fixed dynamical exponent. At any given shell one can isotropize the quadratic form by the scale-dependent rescaling

$$\hat{\varpi} = a_r(\ell) \varpi,$$

but there is no single scale-independent transformation that removes $\tilde{\rho}_r(\varpi)$ at all scales.

The soft dispersion is

$$|\varpi|_\ell \sim \frac{|\tilde{p}|_\ell}{\sqrt{\tilde{\rho}_{r,\ell}}} \sim \frac{|\tilde{p}|_\ell}{\sqrt{\kappa_r \ell}} \sim \frac{|\tilde{p}|_\ell}{\sqrt{\kappa_r \log(\Lambda/|\tilde{p}|_\ell)}}. \quad (205)$$

Thus the leading scaling remains $z = 1$, but with a multiplicative logarithmic correction rather than a pure power law. If one insists on defining a local effective exponent from $|\varpi|_\ell \propto |\tilde{p}|_\ell^{z_{\text{eff}}}$, then

$$z_{\text{eff}}(\ell) = 1 + \frac{\kappa_r}{2\tilde{\rho}_{r,\ell}} \xrightarrow{\ell \rightarrow \infty} 1. \quad (206)$$

So the correct statement is that the aspect ratio runs, but there is no separate one-loop flow of the ballistic velocity or of a universal anisotropic critical exponent.

A. Fermion-Bubble Input to the NLSM

1. Retarded-advanced bubble

The retarded Green's function on the semi-infinite geometry is

$$G_R^{-1}(\omega, q) = (\omega + i\Gamma/2) \mathbb{1} - \sin q \sigma_x - (1 - \cos q) \sigma_z - \Sigma_R(\omega, q) P_-, \quad (A1)$$

with

$$\Sigma_R(\omega, q) = \frac{\omega^2 + 2 \cos q - 1 - \sqrt{\omega^2 - 1 + i0} \sqrt{\omega^2 - (5 - 4 \cos q) + i0}}{2(\omega - \sin q + i0)}. \quad (A2)$$

The retarded-advanced bubble entering the quadratic NLSM is

$$\Pi_{RA}(\Omega, k) = \int \frac{d\omega dq}{(2\pi)^2} \text{tr}_{\text{orb}}[G_R(\omega + \Omega, q + k) G_A(\omega, q)]. \quad (A3)$$

$$\Delta(\Omega, k) = \Gamma \text{Re} \left[\frac{\Pi_{RA}(\Omega, k) + \Pi_{RA}(-\Omega, -k)}{2} - \Pi_{RA}(0, 0) \right]. \quad (A4)$$

For the line-cut benchmarks below we use direct midpoint quadrature on the trusted small windows

$$5 \times 10^{-4} \leq k \leq 2.5 \times 10^{-3}, \quad 5 \times 10^{-4} \leq \Omega \leq 1.5 \times 10^{-3},$$

with datasets `data/smallk_direct_lines_Nw16384_Nq4096_wmax320` and `data/smallomega_direct_lines`

The largest damping used in these fits is $\Gamma = 80$, so the numerical frequency bandwidth always exceeds Γ .

It is also useful to resolve the bubble in the basis where σ_x is diagonal:

$$G'(\omega, q) = \begin{pmatrix} G_{++}(\omega, q) & G_{+-}(\omega, q) \\ G_{-+}(\omega, q) & G_{--}(\omega, q) \end{pmatrix}. \quad (A5)$$

Because the Frobenius overlap is basis-invariant, the RA bubble splits into component-wise contributions of G_{++} , G_{+-} , G_{-+} , and G_{--} . The exact numerical decomposition shows that the low-energy coefficients are dominated by the $--$ channel.

a. Zero-momentum bubble

At zero external arguments,

$$\Pi_{RA}(0, 0) = \int \frac{d\omega}{2\pi} \int_{-\pi}^{\pi} \frac{dq}{2\pi} \sum_{a,b=1}^2 |G_{R,ab}(\omega, q)|^2. \quad (\text{A6})$$

This was evaluated directly with adaptive bandwidth

$$\omega_{\max} = \max(320, 8\Gamma).$$

The raw result has the universal high-frequency tail

$$\Pi_{RA}(0, 0) = \Pi_{\text{finite } \omega_{\max}}(0, 0) + \frac{2}{\pi\omega_{\max}} + o(\omega_{\max}^{-1}),$$

so after tail correction the data collapse rapidly to

$$\Gamma\Pi_{RA}(0, 0) \rightarrow 2. \quad (\text{A7})$$

Numerically, the corrected totals are

$$1.98597, \quad 1.99545, \quad 1.99878, \quad 1.99975, \quad 2.00001$$

for $\Gamma = 10, 20, 40, 80, 160$.

The same limit follows analytically from the large- Γ expansion

$$G_R(\omega, q) = \frac{2}{\Gamma(x+i)} \mathbb{1} + O(\Gamma^{-2}), \quad \omega = \frac{\Gamma x}{2},$$

which gives

$$\Gamma\Pi_{RA}(0, 0) = \int \frac{dq}{2\pi} \int \frac{dx}{\pi} \frac{2}{x^2 + 1} + O(\Gamma^{-1}) = 2 + O(\Gamma^{-1}). \quad (\text{A8})$$

It is useful to spell out this limit more explicitly. Write

$$G_R^{-1}(\omega, q) = (\omega + i\Gamma/2) \mathbb{1} - \mathcal{M}(\omega, q),$$

where

$$\mathcal{M}(\omega, q) = \begin{pmatrix} \sin q & 1 - \cos q \\ 1 - \cos q & -\sin q + \Sigma_R(\omega, q) \end{pmatrix}.$$

For fixed q and scaled frequency $\omega = \Gamma x/2$, the exact bulk self-energy satisfies $\Sigma_R(\Gamma x/2, q) = O(\Gamma^{-1})$ at fixed x , so $\mathcal{M}(\Gamma x/2, q) = O(1)$ uniformly in q . Therefore

$$G_R(\Gamma x/2, q) = \frac{2}{\Gamma} \left[(x+i) \mathbb{1} - \frac{2}{\Gamma} \mathcal{M}(\Gamma x/2, q) \right]^{-1} = \frac{2}{\Gamma(x+i)} \mathbb{1} + O(\Gamma^{-2}).$$

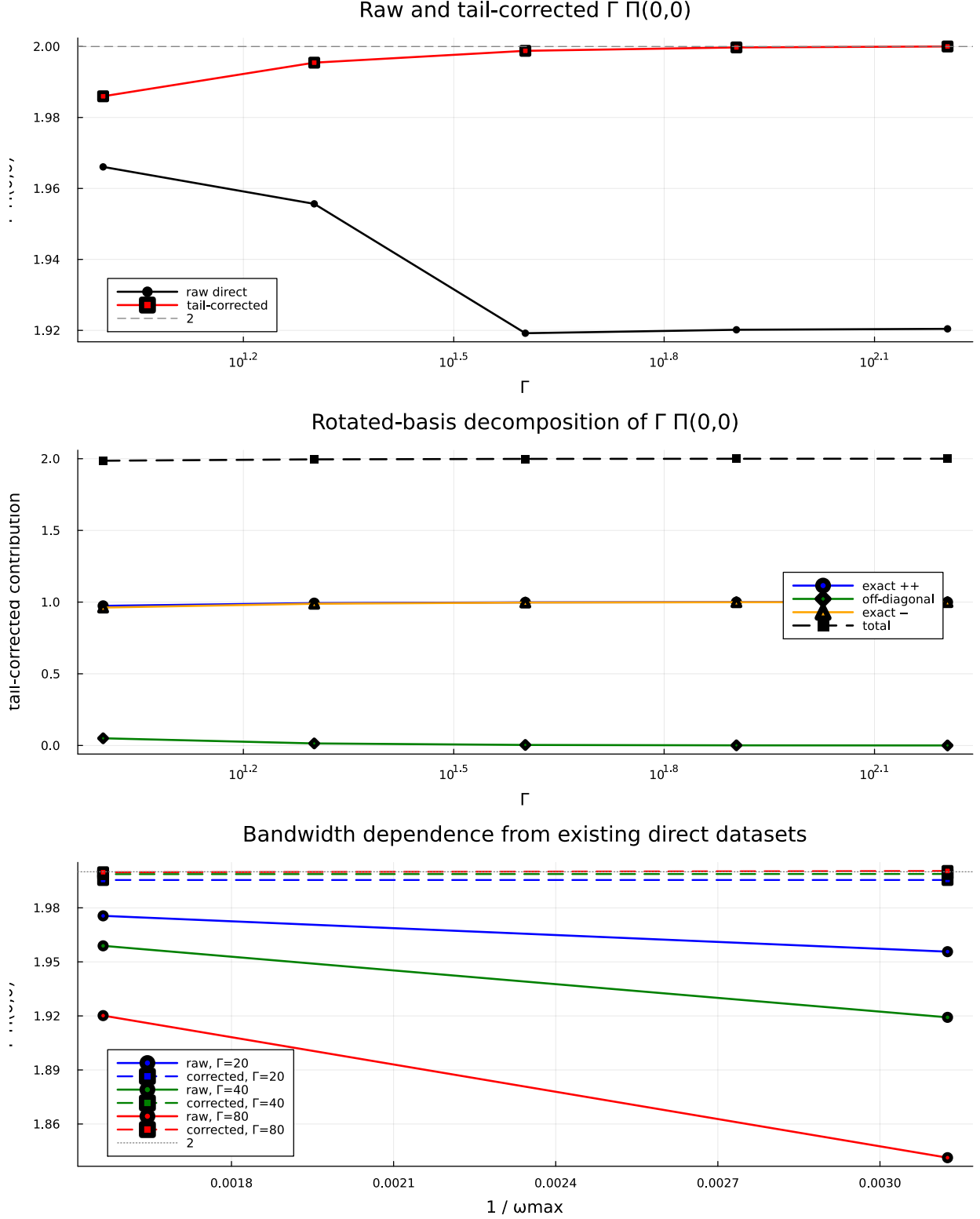


FIG. 1: Direct numerical evaluation of $\Gamma \Pi_{RA}(0,0)$. Top: raw and tail-corrected totals. Middle: decomposition in the σ_x -diagonal basis. Bottom: bandwidth dependence and the corresponding tail correction.

Substituting into (A6) yields

$$\begin{aligned}
\Gamma \Pi_{RA}(0, 0) &= \Gamma \int \frac{dq}{2\pi} \int \frac{d\omega}{2\pi} \text{Tr} [G_R(\omega, q) G_R(\omega, q)^\dagger] \\
&= \Gamma \int \frac{dq}{2\pi} \int \frac{\Gamma dx}{4\pi} \text{Tr} \left[\frac{4}{\Gamma^2(x^2 + 1)} \mathbb{1} + O(\Gamma^{-3}) \right] \\
&= \int \frac{dq}{2\pi} \int \frac{dx}{\pi} \frac{2}{x^2 + 1} + O(\Gamma^{-1}) = 2 + O(\Gamma^{-1}).
\end{aligned} \tag{A9}$$

The $O(\Gamma^{-1})$ remainder above is only a bound from the broad frequency-scaling expansion. The defect in the exact sum rule is smaller. The retarded and advanced Green's functions obey

$$G_R - G_A = G_R \left[(\Sigma_R - \Sigma_A) P_- - i\Gamma \mathbb{1} \right] G_A,$$

so

$$i(G_R - G_A) = \Gamma G_R G_A + i G_R (\Sigma_R - \Sigma_A) P_- G_A.$$

After integrating over frequency and momentum, the spectral sum rule on the left gives 2. Therefore

$$2 - \Gamma \Pi_{RA}(0, 0) = \int \frac{d\omega dq}{(2\pi)^2} [-2 \text{Im} \Sigma_R(\omega, q)] \text{tr}_{\text{orb}} [G_R(\omega, q) P_- G_A(\omega, q)]. \tag{A10}$$

The factor $-2 \text{Im} \Sigma_R$ is supported only on the finite bulk continuum. On that support, $G_R = O(\Gamma^{-1})$, and hence

$$2 - \Gamma \Pi_{RA}(0, 0) = O(\Gamma^{-2}). \tag{A11}$$

There is also a stronger exact statement for the pole-only approximation. Let

$$Z_+(q) = \begin{cases} \cos q (2 - \cos q), & \cos q > 0, \\ 0, & \cos q < 0, \end{cases} \quad \Sigma_{\text{pole}}(z, q) = \frac{Z_+(q)}{z - \sin q},$$

and denote by $G_R^{\text{pole}}(z, q)$ the Green's function obtained by replacing Σ_R with Σ_{pole} . This pole-only Green's function is the physical 2×2 block of the Hermitian 3×3 resolvent

$$\mathcal{G}^{\text{pole}}(z, q) = \left[z \mathbb{1}_3 - \begin{pmatrix} \sin q & 1 - \cos q & 0 \\ 1 - \cos q & -\sin q & \sqrt{Z_+(q)} \\ 0 & \sqrt{Z_+(q)} & \sin q \end{pmatrix} + \frac{i\Gamma}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right]^{-1},$$

namely $G_R^{\text{pole}} = P \mathcal{G}^{\text{pole}} P$ with

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Because the 3×3 Hamiltonian is Hermitian, its resolvent obeys

$$i(\mathcal{G}^{\text{pole}} - \mathcal{G}^{\text{pole}\dagger}) = \Gamma \mathcal{G}^{\text{pole}} P \mathcal{G}^{\text{pole}\dagger}.$$

Projecting onto the physical block gives

$$i(G_R^{\text{pole}} - G_R^{\text{pole}\dagger}) = \Gamma G_R^{\text{pole}} G_R^{\text{pole}\dagger}.$$

Therefore

$$\Gamma \text{Tr} [G_R^{\text{pole}}(\omega, q) G_R^{\text{pole}}(\omega, q)^\dagger] = -2 \text{Im} \text{Tr} G_R^{\text{pole}}(\omega, q).$$

Since $G_R^{\text{pole}}(z, q) = \mathbb{1}_2/z + O(z^{-2})$ as $|z| \rightarrow \infty$, the spectral sum rule gives

$$\int_{-\infty}^{\infty} \frac{d\omega}{\pi} [-\text{Im} \text{Tr} G_R^{\text{pole}}(\omega, q)] = 2$$

for every q . Hence the pole-only zero-momentum bubble is exact:

$$\Gamma \Pi_{RA}^{\text{pole}}(0, 0) = 2. \quad (\text{A12})$$

b. Purely imaginary linear terms

The appendix results used so far concern the symmetrized real part (A4). For the (θ, ϕ) sector one also needs the odd analytic part of the full complex bubble. Expanding the retarded-advanced bubble at small external arguments gives

$$\Pi_{RA}(\Omega, k) - \Pi_{RA}(0, 0) = \Omega \mathcal{I}_\Omega + k \mathcal{I}_k + O(\Omega^2, \Omega k, k^2), \quad (\text{A13})$$

with

$$\mathcal{I}_\Omega \equiv \int \frac{d\omega dq}{(2\pi)^2} \text{tr}_{\text{orb}}[(\partial_\omega G_R)(\omega, q) G_A(\omega, q)], \quad \mathcal{I}_k \equiv \int \frac{d\omega dq}{(2\pi)^2} \text{tr}_{\text{orb}}[(\partial_q G_R)(\omega, q) G_A(\omega, q)]. \quad (\text{A14})$$

Because $G_A(\omega, q) = G_R(\omega, q)^\dagger$ on the real axis,

$$\text{Re} \mathcal{I}_\Omega = \frac{1}{2} \int \frac{d\omega dq}{(2\pi)^2} \partial_\omega \text{tr}_{\text{orb}}[G_R(\omega, q) G_A(\omega, q)] = 0, \quad (\text{A15})$$

$$\text{Re} \mathcal{I}_k = \frac{1}{2} \int \frac{d\omega dq}{(2\pi)^2} \partial_q \text{tr}_{\text{orb}}[G_R(\omega, q) G_A(\omega, q)] = 0, \quad (\text{A16})$$

where the ω total derivative vanishes because $G_R \sim \omega^{-1}$ at large frequency, while the q total derivative vanishes by periodicity. Therefore the linear terms are necessarily pure imaginary:

$$\Pi_{RA}(\Omega, k) - \Pi_{RA}(0, 0) = i \beta_\Omega(\Gamma) \Omega - i \beta_k(\Gamma) k + O(\Omega^2, \Omega k, k^2), \quad (\text{A17})$$

with real coefficients $\beta_\Omega(\Gamma)$ and $\beta_k(\Gamma)$.

The large- Γ scaling of the Ω term can be obtained directly from the leading broadening-dominated Green's function. Writing $\omega = \Gamma x/2$, one has

$$G_R(\omega, q) = \frac{2}{\Gamma(x+i)} \mathbb{1} + O(\Gamma^{-2}), \quad G_A(\omega, q) = \frac{2}{\Gamma(x-i)} \mathbb{1} + O(\Gamma^{-2}).$$

Keeping only this universal leading piece gives

$$\Pi_{RA}(\Omega, 0) = \frac{2}{\Gamma} \int \frac{dx}{\pi} \frac{1}{(x+2\Omega/\Gamma+i)(x-i)} + O(\Gamma^{-2}) = \frac{2}{\Gamma} \frac{1}{1-i\Omega/\Gamma} + O(\Gamma^{-2}), \quad (\text{A18})$$

hence

$$\Gamma [\Pi_{RA}(\Omega, 0) - \Pi_{RA}(0, 0)] = \frac{2i \Omega/\Gamma}{1-i\Omega/\Gamma} + O(\Gamma^{-2}) = \frac{2i \Omega}{\Gamma} + O\left(\frac{\Omega^2}{\Gamma^2}, \Gamma^{-2}\right). \quad (\text{A19})$$

So the odd frequency term is analytic and scales as $1/\Gamma$ in $\Gamma \Pi_{RA}$.

For the momentum term, the same leading broadening-only approximation gives no k dependence at all, because the leading Green's function is proportional to the identity and independent of q . The linear k term therefore first appears through the subleading q -dependent corrections to G_R , so

$$\Gamma \text{Im}[\Pi_{RA}(0, k) - \Pi_{RA}(0, 0)] = O\left(\frac{k}{\Gamma}\right). \quad (\text{A20})$$

Thus both odd linear terms decay with Γ , although the k term is absent in the leading broadening approximation.

A direct recomputation of the full complex line cuts using `src/collect_small_argument_direct_lines` on a moderate grid

$$N_\omega = 4096, \quad N_q = 1024, \quad \omega_{\max} = 320, \quad \eta = 10^{-4},$$

confirms this scaling. At $\Omega = \pm 5 \times 10^{-4}, \pm 10^{-3}$ and $k = \pm 5 \times 10^{-4}, \pm 10^{-3}$, one finds

$$\frac{\Gamma \text{Im}[\Pi_{RA}(\Omega, 0) - \Pi_{RA}(0, 0)]}{\Omega} \approx \frac{2.379}{\Gamma}, \quad \frac{2.377}{\Gamma}, \quad \frac{2.374}{\Gamma}, \quad (\text{A21})$$

$$\frac{\Gamma \text{Im}[\Pi_{RA}(0, k) - \Pi_{RA}(0, 0)]}{k} \approx -\frac{0.313}{\Gamma}, \quad -\frac{0.304}{\Gamma}, \quad -\frac{0.299}{\Gamma}, \quad (\text{A22})$$

for $\Gamma = 20, 40, 80$, respectively. The numerical prefactors are therefore small and decrease approximately as $1/\Gamma$, in agreement with the large- Γ analytic estimate above.

c. Small- k behavior

On the trusted small- k window, the line cut is already very close to a pure quadratic:

$$\text{Re}[\Gamma(\Pi_{RA}(0, k) - \Pi_{RA}(0, 0))] \approx -A_k(\Gamma) k^2.$$

The fitted coefficients are

$$A_k(20) \approx 0.07322, \quad A_k(40) \approx 0.06882, \quad A_k(80) \approx 0.06730,$$

with sub- 10^{-3} relative RMS errors. Figure 2 shows the channel decomposition.

Numerically, the corresponding quadratic coefficients are

Γ	full pole-only	edge exact ++	off-diagonal exact --	
20	0.07322	1.03780	0.00126	0.00249
40	0.06882	1.03729	0.00031	0.00064
80	0.06730	1.03717	0.00008	0.00016

To derive the coefficient analytically, define for any single component

$$\Pi_g(0, k) = \int \frac{d\omega dq}{(2\pi)^2} g(\omega, q) \overline{g(\omega, q - k)}. \quad (\text{A23})$$

Then

$$\frac{\Pi_g(0, k) + \Pi_g(0, -k)}{2} - \Pi_g(0, 0) = -\frac{k^2}{2} \int \frac{d\omega dq}{(2\pi)^2} |\partial_q g(\omega, q)|^2 + O(k^4), \quad (\text{A24})$$

so

$$A_k^{(g)}(\Gamma) = \frac{\Gamma}{2} \int \frac{d\omega dq}{(2\pi)^2} |\partial_q g(\omega, q)|^2. \quad (\text{A25})$$

For the dominant $g = G_{--}$ channel, define

$$Z(q) = \cos q (2 - \cos q).$$

Inside the edge interval $\cos q > 0$, the bulk self-energy has the pole

$$\Sigma_R(\omega, q) = \frac{Z(q)}{\omega - \sin q} + \Sigma_{\text{reg}}(\omega, q).$$

Here “edge interval” means $q \in (-\pi/2, \pi/2)$. For $\cos q < 0$, the numerator of the exact self-energy vanishes at $\omega = \sin q$, so the would-be $1/x$ pole is absent and the leading resonance-strip contribution does not arise. From the matrix in the σ_x -diagonal basis,

$$G_R^{-1}(\omega, q) = \begin{pmatrix} A & -m \\ -m & B \end{pmatrix}, \quad A = \omega - \sin q + i\Gamma/2, \quad m = 1 - \cos q,$$

with

$$B = \omega + \sin q - \Sigma_R(\omega, q) + i\Gamma/2,$$

we have

$$G_{--}(\omega, q) = \frac{1}{B - m^2/A}.$$

At large Γ , $m^2/A = O(\Gamma^{-1})$, so

$$G_{--}(\omega, q) = \frac{1}{\omega + \sin q - \Sigma_R(\omega, q) + i\Gamma/2} + O(\Gamma^{-2}),$$

which is the starting point of the asymptotic analysis.

The full derivative at fixed ω is

$$\partial_q G_{--} = \frac{\partial_q \Sigma_R(\omega, q) - \cos q}{(\omega + \sin q - \Sigma_R(\omega, q) + i\Gamma/2)^2} + O(\Gamma^{-1}),$$

and for the pole part of the self-energy,

$$\partial_q \Sigma_R = \frac{Z'(q)}{\omega - \sin q} + \frac{Z(q) \cos q}{(\omega - \sin q)^2} + \partial_q \Sigma_{\text{reg}}.$$

On the resonance strip

$$\omega - \sin q = O\left(\frac{Z(q)}{\Gamma}\right), \quad \omega + \sin q - \Sigma_R(\omega, q) + i\Gamma/2 = O(\Gamma).$$

The four numerator terms then scale as

$$\cos q = O(1), \quad \partial_q \Sigma_{\text{reg}} = O(1), \quad \frac{Z'(q)}{\omega - \sin q} = O(\Gamma), \quad \frac{Z(q) \cos q}{(\omega - \sin q)^2} = O(\Gamma^2).$$

After division by the $O(\Gamma^2)$ denominator in $\partial_q G_{--}$, only the last term survives at leading order. This is why the explicit derivative of $+\sin q$ and the regular part of the self-energy are subleading in the large- Γ asymptotics.

Therefore the leading contribution to the gradient formula can be written directly in the original variables as

$$A_k^{(--)}(\Gamma) \sim \frac{\Gamma}{2} \int_{-\pi/2}^{\pi/2} \frac{dq}{2\pi} \int \frac{d\omega}{2\pi} \frac{Z(q)^2 \cos^2 q}{(\omega - \sin q)^4 \left| \omega + \sin q - \frac{Z(q)}{\omega - \sin q} + i\Gamma/2 \right|^4}. \quad (\text{A26})$$

This integral is concentrated on the narrow strip

$$\omega - \sin q = O\left(\frac{Z(q)}{\Gamma}\right),$$

so it is convenient only at this stage to set

$$\omega - \sin q = \frac{Z(q)}{\Gamma} u.$$

Then

$$\omega + \sin q - \frac{Z(q)}{\omega - \sin q} + i\Gamma/2 = \Gamma \left(\frac{i}{2} - \frac{1}{u} \right) + O(1),$$

and

$$\partial_q G_{--} = \frac{4 \cos q}{Z(q)(iu - 2)^2} + O(\Gamma^{-1}).$$

Substituting back into the gradient formula yields

$$A_k^{(--)}(\infty) = \frac{1}{8\pi} \int_{-\pi/2}^{\pi/2} \frac{\cos q}{2 - \cos q} dq = \frac{1}{3\sqrt{3}} - \frac{1}{8} \approx 0.06745009. \quad (\text{A27})$$

This agrees very well with the extracted exact--- coefficients above, so the full small- k coefficient inherits the same limit.

d. Small- Ω behavior

On the trusted small- Ω window,

$$\text{Re}[\Gamma(\Pi_{RA}(\Omega, 0) - \Pi_{RA}(0, 0))] \approx -\hat{A}_\Omega(\Gamma) \Omega^2.$$

The fitted full coefficients are

$$\hat{A}_\Omega(20) \approx 0.38406, \quad \hat{A}_\Omega(40) \approx 0.35505, \quad \hat{A}_\Omega(80) \approx 0.33140.$$

Again the exact -- channel dominates, as shown in Fig. 3.

The fitted coefficients are

Γ	full	exact ++	exact +-	exact -+	exact --
20	0.38406	-0.00173	0.00419	0.00419	0.37742
40	0.35505	0.00032	0.00066	0.00066	0.35341
80	0.33140	0.00014	0.00013	0.00013	0.33101

For a single component,

$$\frac{\Pi_g(\Omega, 0) + \Pi_g(-\Omega, 0)}{2} - \Pi_g(0, 0) = -\frac{\Omega^2}{2} \int \frac{d\omega dq}{(2\pi)^2} |\partial_\omega g(\omega, q)|^2 + O(\Omega^4), \quad (\text{A28})$$

so

$$A_{\Omega}^{(g)}(\Gamma) = \frac{\Gamma}{2} \int \frac{d\omega dq}{(2\pi)^2} |\partial_{\omega} g(\omega, q)|^2. \quad (\text{A29})$$

For $g = G_{--}$, the same large- Γ form

$$G_{--}(\omega, q) = \frac{1}{\omega + \sin q - \Sigma_R(\omega, q) + i\Gamma/2} + O(\Gamma^{-2})$$

still applies. On the resonance strip

$$\omega - \sin q = \frac{Z(q)}{\Gamma} u,$$

the pole part of the self-energy gives

$$\partial_{\omega} \Sigma_R \sim -\frac{Z(q)}{(\omega - \sin q)^2}, \quad \partial_{\omega} G_{--} = \frac{4}{Z(q)(iu - 2)^2} + O(\Gamma^{-1}).$$

Substituting into the gradient formula gives the formal density

$$A_{\Omega}^{(--)}(\Gamma) \sim \frac{1}{8\pi} \int_{-\pi/2}^{\pi/2} \frac{dq}{Z(q)},$$

which diverges logarithmically at the touching points because $Z(q) \rightarrow 0$ as $q \rightarrow \pm\pi/2$.

Keeping $\Omega \neq 0$ as the infrared regulator yields

$$\text{Re}[\Gamma(\Pi_{RA,--}(\Omega, 0) - \Pi_{RA,--}(0, 0))]_{\text{sing}} = -\frac{\Omega^2}{8\pi} \log \frac{C_{\Omega}}{\Gamma|\Omega|} + O(\Omega^2). \quad (\text{A30})$$

This does not directly predict the finite-window fitted coefficient $\hat{A}_{\Omega}(\Gamma)$, but it does imply the drift

$$\hat{A}_{\Omega,--}(\Gamma) = B_{\Omega} - \frac{1}{8\pi} \log \Gamma + o(1).$$

Numerically, the shifted combination

$$\hat{A}_{\Omega,--}(\Gamma) + \frac{1}{8\pi} \log \Gamma$$

is nearly constant:

$$0.49661, \quad 0.50018, \quad 0.50536 \quad (\Gamma = 20, 40, 80),$$

which is the appropriate benchmark of the logarithmic prediction.

e. Mixed Ωk term

To isolate the bilinear term, define the odd-odd projection

$$\Delta_{\text{odd-odd}}(\Omega, k) = \frac{\Gamma \text{Re}\Pi_{RA}(\Omega, k) - \Gamma \text{Re}\Pi_{RA}(\Omega, -k) - \Gamma \text{Re}\Pi_{RA}(-\Omega, k) + \Gamma \text{Re}\Pi_{RA}(-\Omega, -k)}{4}. \quad (\text{A31})$$

For small arguments,

$$\Delta_{\text{odd-odd}}(\Omega, k) = A_{\Omega k}(\Gamma) \Omega k + o(\Omega k).$$

The direct decomposition shows that this coefficient is also overwhelmingly the exact $--$ channel. At the small point

$$\Omega = k = 5 \times 10^{-4}, \quad N_\omega = 16384, \quad N_q = 4096, \quad \omega_{\text{max}} = 640,$$

the odd-odd quotient gives

Γ	full exact	++ off-diagonal	exact	--
20	0.19760	0.00011	0.00074	0.19675
40	0.19283	0.00001	0.00016	0.19266
80	0.19038	0.00000	0.00004	0.19034

At the still smaller point $\Omega = k = 2.5 \times 10^{-4}$, one finds

$$A_{\Omega k}^{\text{full}}(80) \approx 0.19130, \quad A_{\Omega k}^{(--)}(80) \approx 0.19126.$$

For a single component,

$$A_{\Omega k}^{(g)}(\Gamma) = -\Gamma \text{Re} \int \frac{d\omega dq}{(2\pi)^2} (\partial_\omega g)(\partial_q \bar{g}). \quad (\text{A32})$$

Applying this to $g = G_{--}$, and using

$$\partial_\omega G_{--} = -\frac{4}{Z(q)(iu-2)^2} + O(\Gamma^{-1}), \quad \partial_q G_{--} = \frac{4 \cos q}{Z(q)(iu-2)^2} + O(\Gamma^{-1}),$$

on the same resonance strip, one obtains

$$A_{\Omega k}^{(--)}(\infty) = \frac{1}{4\pi} \int_{-\pi/2}^{\pi/2} \frac{dq}{2 - \cos q} = \frac{1}{3\sqrt{3}} \approx 0.19245009. \quad (\text{A33})$$

This matches the direct high-resolution values above very well.

Collecting the results,

$$\Delta(\Omega, k) = -A_k k^2 + A_{\Omega k} \Omega k - \frac{\Omega^2}{8\pi} \log \frac{C_\Omega}{\Gamma|\Omega|} + \dots, \quad (\text{A34})$$

with

$$A_k \xrightarrow{\Gamma \rightarrow \infty} \frac{1}{3\sqrt{3}} - \frac{1}{8}, \quad A_{\Omega k} \xrightarrow{\Gamma \rightarrow \infty} \frac{1}{3\sqrt{3}}, \quad \Gamma \Pi_{RA}(0, 0) \xrightarrow{\Gamma \rightarrow \infty} 2. \quad (\text{A35})$$

These are precisely the coefficients used in the main text.

f. Reduced G_{--} bubble and direct kernel expansion

Within the leading large- Γ resonance-strip approximation, the $--$ Green's function can be replaced by the local model

$$G_{--}^{\text{loc}}(\omega, q) = \frac{\omega - \sin q}{i\Gamma(\omega - \sin q)/2 - Z(q)}, \quad Z(q) = \cos q (2 - \cos q), \quad (\text{A36})$$

for $q \in (-\pi/2, \pi/2)$. This is not the full lattice Green's function; it is the leading edge-strip model that keeps only the pole part of the bulk self-energy.

Whenever both q and $q - k$ remain inside the edge interval, the local $--$ bubble is

$$\Pi_{RA,--}^{\text{loc}}(\Omega, k) = \int_{-\pi/2}^{\pi/2} \frac{dq}{2\pi} \int \frac{d\omega}{2\pi} G_{--}^{\text{loc}}(\omega + \Omega, q + k) \overline{G_{--}^{\text{loc}}(\omega, q)}.$$

The ω integral can be done explicitly by residues. Closing the contour in the upper half-plane picks the pole of the advanced factor and gives

$$\Pi_{RA,--}^{\text{loc}}(\Omega, k) = \int_{-\pi/2}^{\pi/2} \frac{dq}{2\pi} \mathcal{I}_\Gamma(q; \Omega, k), \quad (\text{A37})$$

with

$$\mathcal{I}_\Gamma(q; \Omega, k) = \frac{4Z(q)}{\Gamma^2} \frac{i(\Omega + \sin q - \sin(q + k)) - 2Z(q)/\Gamma}{Z(q) + Z(q + k) - i\Gamma(\Omega + \sin q - \sin(q + k))/2}. \quad (\text{A38})$$

This is a useful one-dimensional summary of the reduced $--$ bubble.

To expand it directly, first take its real part. One finds

$$\Gamma \text{Re} \mathcal{I}_\Gamma(q; \Omega, k) = - \frac{2Z(q)(\Omega + \sin q - \sin(q + k))^2 + \frac{8}{\Gamma^2} Z(q)^2 (Z(q) + Z(q + k))}{(Z(q) + Z(q + k))^2 + \frac{\Gamma^2}{4} (\Omega + \sin q - \sin(q + k))^2}. \quad (\text{A39})$$

For small arguments,

$$Z(q + k) = Z(q) + O(k), \quad \Omega + \sin q - \sin(q + k) = \Omega - k \cos q + O(k^2),$$

so

$$\Gamma \operatorname{Re} \mathcal{I}_\Gamma(q; \Omega, k) = -\frac{4Z(q)}{\Gamma^2} - \frac{(\Omega - k \cos q)^2}{4Z(q)} + O\left(\frac{k}{\Gamma^2}, \frac{\Omega^2 + k^2}{\Gamma^2}, |\Omega|^3, |\Omega|^2|k|, |\Omega|k^2, |k|^3\right).$$

Therefore,

$$\begin{aligned} \frac{\Gamma \operatorname{Re} \mathcal{I}_\Gamma(q; 0, k) + \Gamma \operatorname{Re} \mathcal{I}_\Gamma(q; 0, -k)}{2} - \Gamma \operatorname{Re} \mathcal{I}_\Gamma(q; 0, 0) &= -\frac{\cos^2 q}{4Z(q)} k^2 + O\left(k^4, \frac{k^2}{\Gamma^2}\right), \\ \frac{\Gamma \operatorname{Re} \mathcal{I}_\Gamma(q; \Omega, k) - \Gamma \operatorname{Re} \mathcal{I}_\Gamma(q; \Omega, -k) - \Gamma \operatorname{Re} \mathcal{I}_\Gamma(q; -\Omega, k) + \Gamma \operatorname{Re} \mathcal{I}_\Gamma(q; -\Omega, -k)}{4} &= \frac{\cos q}{2Z(q)} \Omega k + o(\Omega k), \\ \frac{\Gamma \operatorname{Re} \mathcal{I}_\Gamma(q; \Omega, 0) + \Gamma \operatorname{Re} \mathcal{I}_\Gamma(q; -\Omega, 0)}{2} - \Gamma \operatorname{Re} \mathcal{I}_\Gamma(q; 0, 0) &= -\frac{\Omega^2}{4Z(q)} + O\left(\Omega^4, \frac{\Omega^2}{\Gamma^2}\right). \end{aligned} \tag{A40}$$

After integrating over q with measure $dq/(2\pi)$, these direct kernel expansions reproduce exactly the coefficient densities used above:

$$\frac{\cos^2 q}{8\pi Z(q)} = \frac{\cos q}{8\pi(2 - \cos q)}, \quad \frac{\cos q}{4\pi Z(q)} = \frac{1}{4\pi(2 - \cos q)}, \quad \frac{1}{8\pi Z(q)}.$$

The first two integrate to (A27) and (A33). The third is not integrable at $q = \pm\pi/2$, which is the direct kernel-level origin of the logarithm in (A30).

2. Large- ΓQ endpoint crossover

The coefficient extraction in Sec. VI is a small- (Ω, k) expansion at fixed Γ . The same reduced endpoint kernel has a different leading form when Γ is taken large while the external arguments are held fixed relative to $1/\Gamma$. This is the origin of the nonuniformity discussed in the main text.

For generic $0 < r < 2$, write

$$q_r = \arccos(r - 1), \quad c_r = r - 1, \quad s_r = \sqrt{r(2 - r)},$$

$$Z_r(q) = 1 - (r - \cos q)^2, \quad D(q) = \Omega - k \cos q.$$

The reduced endpoint kernel is

$$\Delta_{\text{end}}(\Omega, k) = - \int_{-q_r}^{q_r} \frac{dq}{2\pi} \frac{Z_r(q) D(q)^2}{4Z_r(q)^2 + \Gamma^2 D(q)^2/4}. \tag{A41}$$

The Taylor/log regime follows by expanding this expression when $\Gamma|D(q)| \ll Z_r(q)$ away from the endpoint, with the endpoint itself regulated by $\Gamma|\Omega - c_r k|$. In contrast, when

$\Gamma Q \gg 1$ and $D(q)$ is not parametrically small over a finite interval, the broadening term dominates the denominator over almost all q . Then

$$\frac{Z_r(q)D(q)^2}{4Z_r(q)^2 + \Gamma^2 D(q)^2/4} = \frac{4Z_r(q)}{\Gamma^2} + \dots, \quad (\text{A42})$$

and the leading endpoint contribution saturates to

$$\Delta_{\text{end}}(\Omega, k) = -\frac{2}{\pi\Gamma^2}I_Z(r) + \dots, \quad I_Z(r) \equiv \int_{-q_r}^{q_r} dq Z_r(q). \quad (\text{A43})$$

The remaining integral is elementary:

$$I_Z(r) = (1 - 2r^2)q_r + (3r + 1)s_r. \quad (\text{A44})$$

Thus the leading $\Gamma Q \gg 1$ endpoint term is independent of the direction from which (Ω, k) approaches the origin, as long as the approach is kept outside parametrically small crossover regions. If $D(q)$ has an interior zero, the zero only produces a subleading correction because the numerator also contains $D(q)^2$.

The tangent direction $\Omega = c_r k$ is the most delicate case because the endpoint mismatch itself vanishes. It is nevertheless described by the same large- ΓQ saturation. Setting $x(q) = \cos q - c_r$ and $A(q) = r + 1 - \cos q$, so that $Z_r(q) = x(q)A(q)$, one has

$$D(q) = -kx(q) \quad (\Omega = c_r k).$$

Substituting this into (A41) gives the exact reduced endpoint expression

$$\Delta_{\text{tan}}(k; \Gamma) = -\frac{k^2}{2\pi} \int_{-q_r}^{q_r} dq \frac{x(q)A(q)}{4A(q)^2 + \Gamma^2 k^2/4}. \quad (\text{A45})$$

For $\Gamma|k| \ll 1$, this reduces to the regular Taylor result,

$$\Delta_{\text{tan}}(k; \Gamma) = \frac{q_r - J_r}{4\pi} k^2 + \dots, \quad (\text{A46})$$

which is the value obtained from $\Delta_{r,\text{reg}}^{(--)}(c_r k, k)$. For $\Gamma|k| \gg 1$, however, (A45) gives

$$\Delta_{\text{tan}}(k; \Gamma) = -\frac{2}{\pi\Gamma^2}I_Z(r) + O\left(\frac{1}{\Gamma^4 k^2}\right). \quad (\text{A47})$$

Therefore the limits do not commute:

$$\lim_{\Gamma \rightarrow \infty} \lim_{Q \rightarrow 0} \Delta_{\text{end}}(\Omega, k) \neq \lim_{Q \rightarrow 0} \lim_{\Gamma \rightarrow \infty} \Delta_{\text{end}}(\Omega, k).$$

The coefficient formulas in Sec. VI refer to the first ordering, or equivalently to the regime $\Gamma Q \ll 1$. The saturation formula (A43) describes the opposite regime $\Gamma Q \gg 1$.

We verified this crossover numerically using `src/verify_generic_r_regular_terms.jl`, which directly integrates (A41). The table uses $\Gamma = 2000$ and $k = 0.02$. The tangent column sets $\Omega = c_r k$, while the off-tangent column uses $\Omega = 0.03$ and $k = 0.02$. The ratios compare the numerical integral to $\Delta_{\text{sat}} = -2I_Z(r)/(\pi\Gamma^2)$.

r	Δ_{sat}	$\Delta_{\text{tan}}/\Delta_{\text{sat}}$	$\Delta_{\text{off}}/\Delta_{\text{sat}}$
0.5	-5.11247×10^{-7}	0.98900	0.98831
1.0	-3.86620×10^{-7}	0.98395	0.98000
1.5	-1.74744×10^{-7}	0.97461	0.98717

The percent-level deviations are the expected finite- ΓQ corrections; they decrease as ΓQ is increased.

a. Full retarded-advanced Green's function

The preceding computation isolates the nonuniform endpoint part of the $--$ channel. For the full retarded-advanced bubble, the remaining $++$, $+-$, $-+$, and regular $--$ pieces also contribute analytic large- Γ terms. It is useful to keep the complex, unsymmetrized subtracted kernel

$$K_{RA}(\Omega, k) \equiv \Gamma [\Pi_{RA}(\Omega, k) - \Pi_{RA}(0, 0)]. \quad (\text{A48})$$

The diagonal real kernel used in (A4) is $\text{Re}[K_{RA}(\Omega, k) + K_{RA}(-\Omega, -k)]/2$, while the odd θ - ϕ kernel is controlled by the odd imaginary part. Therefore the linear terms in K_{RA} should be kept even though they drop out of Δ .

Let

$$h_r(q) = \sin q \sigma_x + (r - \cos q) \sigma_z. \quad (\text{A49})$$

The monitored self-energy is $O(\Gamma^{-1})$ for $\omega = \Gamma x/2$, so to the order needed here

$$G_R\left(\frac{\Gamma x}{2} + \Omega, q + k\right) = \frac{2}{\Gamma} \frac{1}{x + i + 2\Omega/\Gamma} + \frac{4}{\Gamma^2} \frac{h_r(q + k)}{(x + i)^2} + O(\Gamma^{-3}), \quad (\text{A50})$$

and similarly for G_A . The scalar part gives exactly

$$K_0(\Omega) = \frac{2}{1 - i\Omega/\Gamma} - 2 = \frac{2i\Omega}{\Gamma} - \frac{2\Omega^2}{\Gamma^2} + O\left(\frac{\Omega^3}{\Gamma^3}\right). \quad (\text{A51})$$

The $h_r h_r$ term is the first part with momentum dependence. At $\Omega = 0$,

$$\begin{aligned} K_h(0, k) &= \frac{2}{\Gamma^2} \int_{-\pi}^{\pi} \frac{dq}{2\pi} \text{tr} [h_r(q+k)h_r(q) - h_r(q)^2] + O(\Gamma^{-3}) \\ &= \frac{4(\cos k - 1)}{\Gamma^2} + O(\Gamma^{-3}) = -\frac{2k^2}{\Gamma^2} + O\left(\frac{k^4}{\Gamma^2}, \Gamma^{-3}\right). \end{aligned} \quad (\text{A52})$$

The trace average has no term linear in k . Since the scalar denominator is independent of q , this also implies that no Ωk term is produced at the displayed order. Possible odd-in- k pieces of the full numerical kernel are therefore subleading endpoint or self-energy corrections, not second-order coefficients of the broadening-dominated full Green's function.

Combining the smooth full-Green contribution with the endpoint saturation (A43), the full complex RA kernel in the regime

$$Q \ll 1, \quad \Gamma Q \gg 1,$$

is

$$K_{RA}(\Omega, k) = \frac{2i\Omega}{\Gamma} - \frac{2I_Z(r)/\pi + 2\Omega^2 + 2k^2}{\Gamma^2} + R_{\text{ep}}(\Omega, k; \Gamma) + \dots \quad (\text{A53})$$

Here R_{ep} denotes the direction-dependent boundary-layer correction left after subtracting the endpoint saturation. If $D(q) = \Omega - k \cos q$ is bounded away from zero on the endpoint interval, then $R_{\text{ep}} = O(\Gamma^{-4}Q^{-2})$. A simple interior zero of D gives a larger but still subleading $O(\Gamma^{-3}Q^{-1})$ correction, and the tangent endpoint case gives the $O(\Gamma^{-4}k^{-2})$ correction in (A47). These terms can be larger than the smooth Q^2/Γ^2 correction for some paths to the origin, but they are smaller than the leading Γ^{-2} saturation whenever $\Gamma Q \gg 1$.

Equation (A53) contains the requested polynomial terms through second order in the external arguments: a linear imaginary Ω term, no leading linear k term, quadratic Ω^2 and k^2 terms, and no Ωk term at this order.

We verified (A53) by direct quadrature of the full orbital trace in (41), using `src/verify_full_large_gammaQ.jl`. The script uses a composite frequency grid, dense near the edge band and coarse in the high-frequency tails. The table compares the numerical value of K_{RA} to the displayed asymptote.

r	Γ	Ω	k	$\text{Re}K_{\text{num}}$	$\text{Re}K_{\text{asy}}$	$\text{Im}K_{\text{num}}$	$\text{Im}K_{\text{asy}}$
0.5	200	0.100	0	-4.93431×10^{-5}	-5.16247×10^{-5}	1.00794×10^{-3}	1.00000×10^{-3}
1.0	200	0.100	0	-3.73623×10^{-5}	-3.91620×10^{-5}	1.00622×10^{-3}	1.00000×10^{-3}
1.5	200	0.100	0	-1.73166×10^{-5}	-1.79744×10^{-5}	1.00186×10^{-3}	1.00000×10^{-3}
1.0	200	0	0.100	-3.63248×10^{-5}	-3.91620×10^{-5}	-9.23737×10^{-6}	0
1.0	400	0	0.100	-9.39823×10^{-6}	-9.79049×10^{-6}	-1.21641×10^{-6}	0
1.5	200	0.100	0.050	-1.67254×10^{-5}	-1.80994×10^{-5}	1.00337×10^{-3}	1.00000×10^{-3}

The pure- Ω rows confirm the leading imaginary term $2\Omega/\Gamma$ and the real Ω^2/Γ^2 correction. The pure k rows test the real k^2/Γ^2 correction together with the endpoint saturation. The small residual imaginary value in those rows is odd in k and falls by nearly a factor of eight when Γ is doubled from 200 to 400, consistent with a subleading remainder rather than a leading linear k term. The mixed row checks that no Ωk coefficient is needed at this order.

3. Full Keldysh-Keldysh bubble

For the full Green's function used in the numerics, the Keldysh component is not obtained from a pole-only ansatz. Instead, the upper-triangular Dyson construction gives

$$G_K(\omega, q) = \text{sgn}(\omega) \left[G_R(\omega, q) - G_A(\omega, q) + i\Gamma G_R(\omega, q) G_A(\omega, q) \right]. \quad (\text{A54})$$

Using

$$G_R - G_A = G_R \left[(\Sigma_R - \Sigma_A) P_- - i\Gamma \mathbb{1} \right] G_A, \quad (\text{A55})$$

this can be rewritten exactly as

$$G_K(\omega, q) = \text{sgn}(\omega) G_R(\omega, q) (\Sigma_R - \Sigma_A)(\omega, q) P_- G_A(\omega, q). \quad (\text{A56})$$

This form makes the large- Γ behavior transparent. Since

$$G_R(\omega, q) = -\frac{2i}{\Gamma} \mathbb{1} + O(\Gamma^{-2}), \quad G_A(\omega, q) = \frac{2i}{\Gamma} \mathbb{1} + O(\Gamma^{-2}), \quad (\text{A57})$$

we obtain

$$G_K(\omega, q) = \frac{4}{\Gamma^2} \text{sgn}(\omega) (\Sigma_R - \Sigma_A)(\omega, q) P_- + O(\Gamma^{-3}). \quad (\text{A58})$$

Therefore

$$\Pi_{KK}(\Omega, k) = \frac{16}{\Gamma^4} \int \frac{d\omega dq}{(2\pi)^2} \text{sgn}(\omega + \Omega) \text{sgn}(\omega) \left[\Sigma_R - \Sigma_A \right](\omega + \Omega, q + k) \left[\Sigma_R - \Sigma_A \right](\omega, q) + O(\Gamma^{-5}). \quad (\text{A59})$$

At $r = 1$, the bulk continuum starts at $|\omega| = 1$, so $\Sigma_R - \Sigma_A$ has no support near $\omega = 0$. Hence the full KK bubble is smooth near $(\Omega, k) = (0, 0)$. The singular shell found in the strict pole-only, zero-temperature approximation is therefore absent once the continuum part of the self-energy is kept.

This conclusion is confirmed numerically. Using direct midpoint quadrature with

$$N_\omega = 8192, \quad N_q = 512, \quad \omega_{\max} = 640, \quad (\text{A60})$$

we find

$$\text{Re}\Pi_{KK}(0, 0) \approx -8.38 \times 10^{-5}, \quad -8.08 \times 10^{-6}, \quad -6.70 \times 10^{-7} \quad \text{for } \Gamma = 20, 40, 80. \quad (\text{A61})$$

On the window $0 \leq \Omega, k \leq 0.04$, the largest resolved line-cut deviations are

$$\max |\text{Re}[\Pi_{KK}(\Omega, 0) - \Pi_{KK}(0, 0)]| \approx 5.23 \times 10^{-6}, \quad 9.62 \times 10^{-7}, \quad 1.21 \times 10^{-7}, \quad (\text{A62})$$

$$\max |\text{Re}[\Pi_{KK}(0, k) - \Pi_{KK}(0, 0)]| \approx 1.24 \times 10^{-7}, \quad 1.71 \times 10^{-8}, \quad 2.21 \times 10^{-9}. \quad (\text{A63})$$

The line cuts and two-dimensional low-energy maps are shown in Figs. 4 and 5. They are smooth at the origin and decay rapidly with Γ , consistent with (A59).

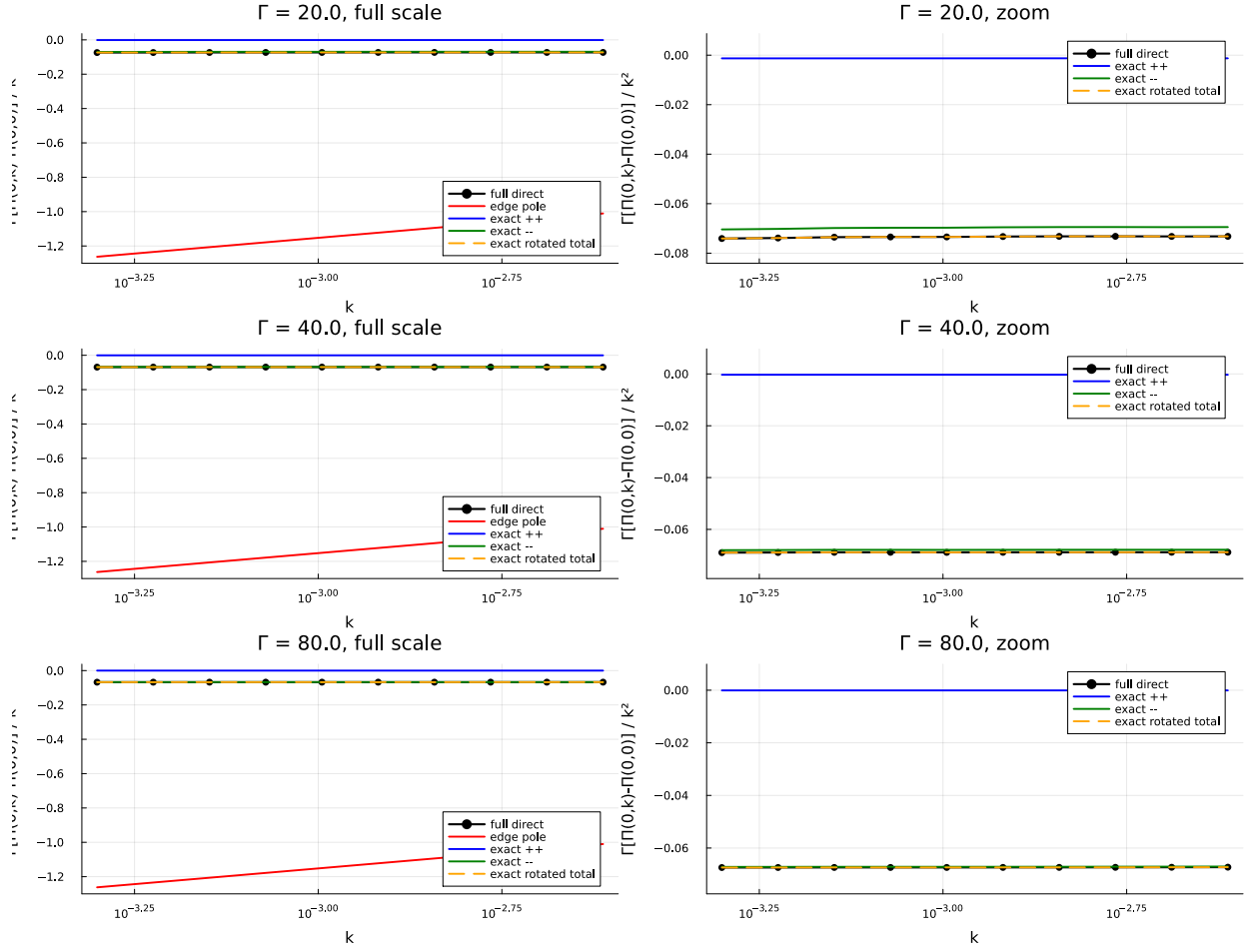


FIG. 2: Decomposition of the small- k line in the σ_x -diagonal basis. The pole-only $++$ edge estimate fails badly, while the exact $--$ channel nearly saturates the full direct result.

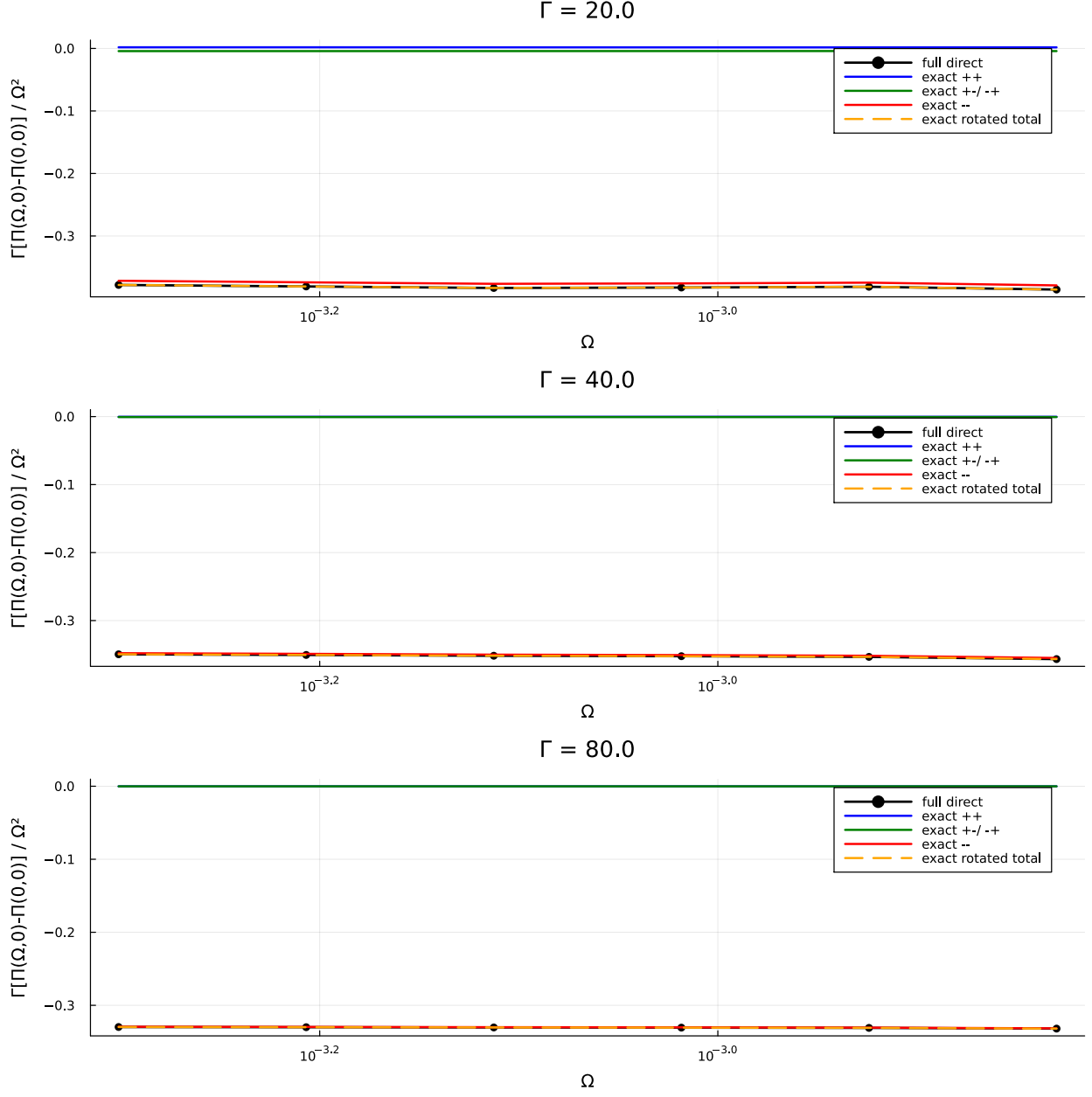


FIG. 3: Decomposition of the small- Ω line in the σ_x -diagonal basis. The exact $--$ contribution already saturates the full direct line for $\Gamma = 40$ and 80 .

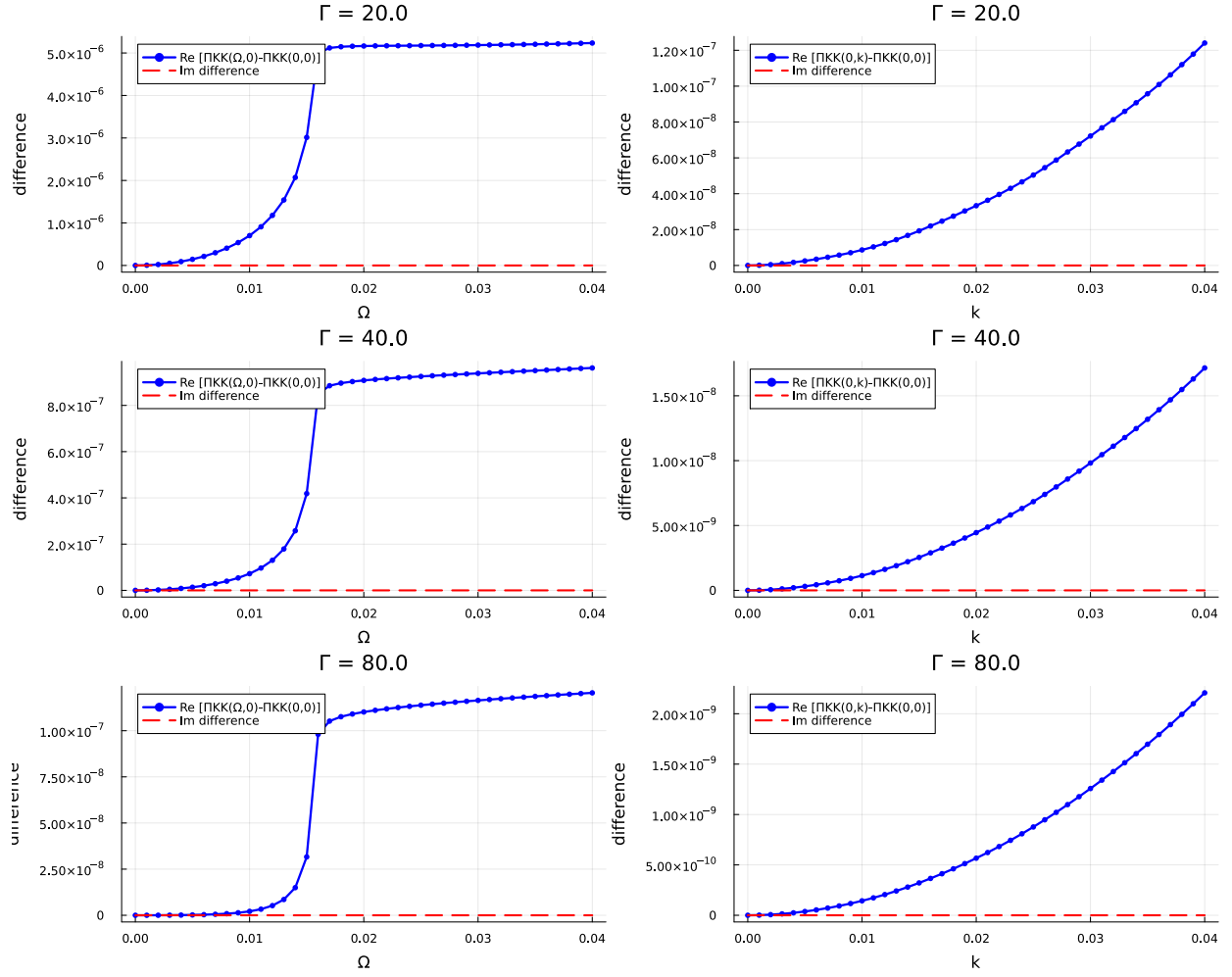


FIG. 4: Direct numerical line cuts of the full KK bubble for $\Gamma = 20, 40, 80$. Left: $\Pi_{KK}(\Omega, 0) - \Pi_{KK}(0, 0)$. Right: $\Pi_{KK}(0, k) - \Pi_{KK}(0, 0)$.

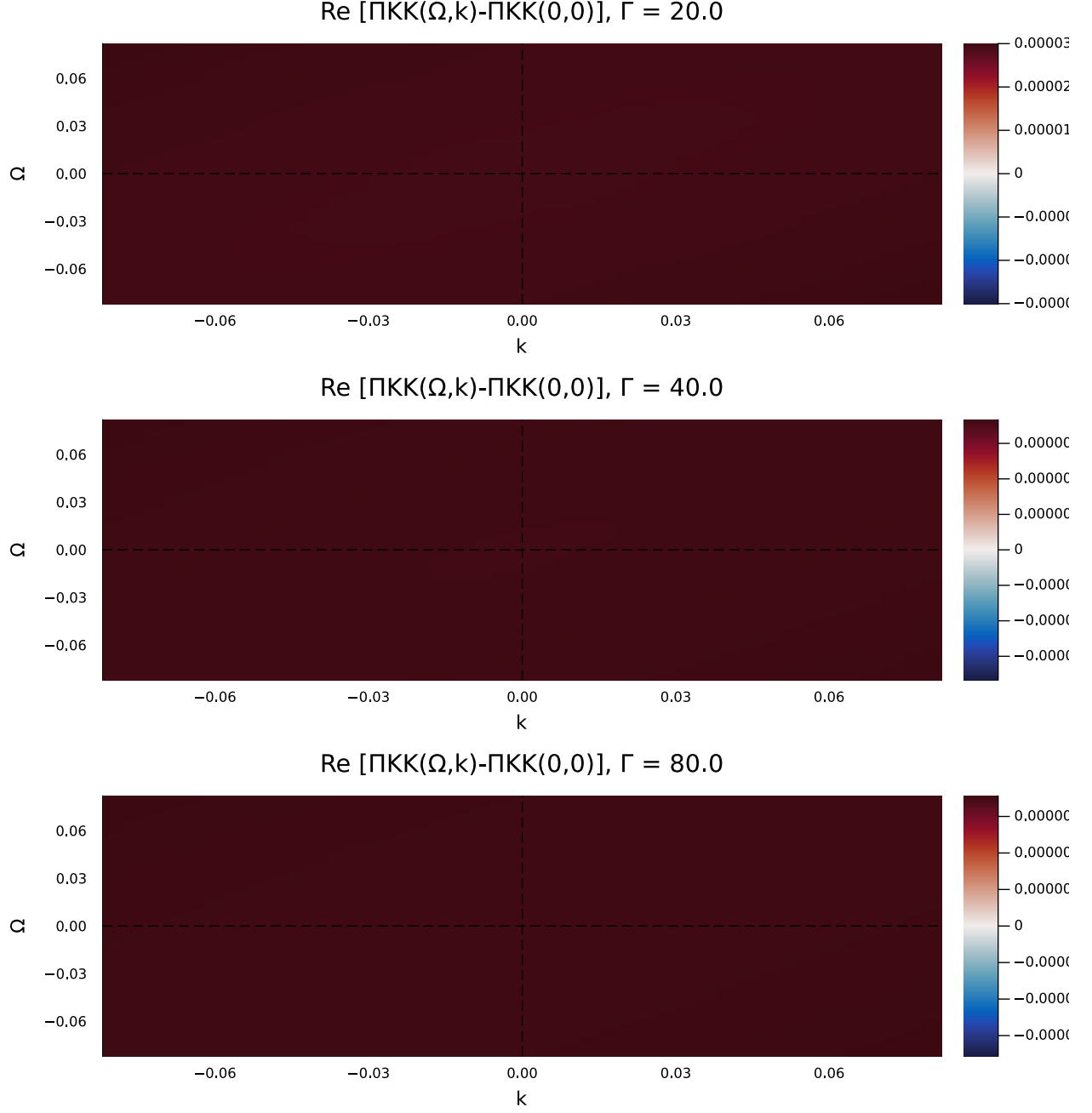


FIG. 5: Direct numerical low-energy maps of $\text{Re}[\Pi_{KK}(\Omega, k) - \Pi_{KK}(0, 0)]$ for the full Green's function. The origin is smooth, and no wedge-like singular support remains.